



Interfacial mechanisms and microstructural evolution in nanosecond laser transmission welding of stainless steel and quartz glass



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Highlights

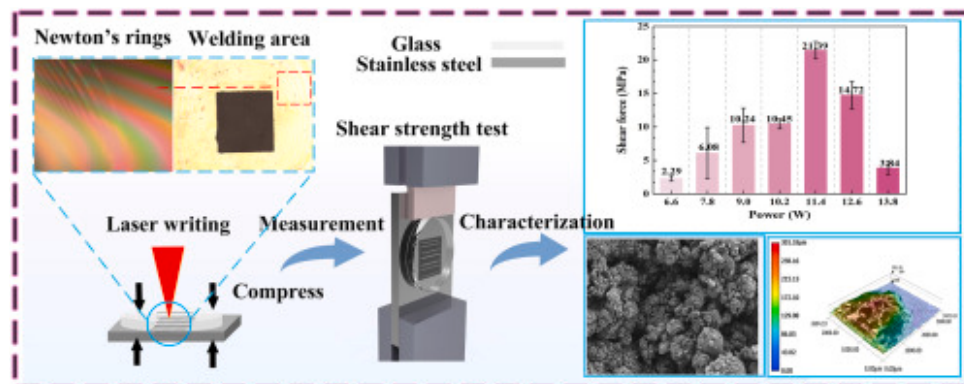
- Nanosecond laser welding joins glass and steel using external pressure.
- Peak shear strength of 21.41 MPa is achieved with optimized parameters.
- Weld zone includes smooth, fragment, and ablation regions.

- SEM, EDS, and XPS reveal interlocking, melting, and chemical bonding.

Abstract

This study investigates the interfacial bonding mechanisms and morphological evolution in nanosecond laser transmission welding of 304 stainless steel and quartz glass under controlled optical contact conditions. A series of experiments were conducted to examine how variations in laser power, pulse frequency, scanning speed, and scanning cycles influence joint formation and bonding behavior. Scanning electron microscopy and energy-dispersive spectroscopy (EDS) were employed to characterize the microstructural features and elemental distribution at the weld interface. The welded joints exhibited three distinct morphological regions: smooth, fragmented, and ablated. Among these regions, the smooth zone contributed most significantly to the integrity of the bonding interface. EDS mapping and morphological observations suggest that localized melting and mechanical interlocking act synergistically during joint formation. Localized melting initiates material mixing and promotes the formation of interlocking features, which collectively contributes to high joint strength. Upon cooling, a dense composite interface is formed, with fracture predominantly occurring within the glass layer, indicating cohesive failure. High-resolution X-ray photoelectron spectroscopy revealed the formation of interfacial Me-O-Si (metal-oxygen-silicon) bonds, particularly Fe-O-Si, which provide strong covalent linkages across the interface. This chemical bonding mechanism further reinforces the joint beyond physical interlocking, contributing to the high strength observed. These findings provide new insights into the interfacial dynamics and laser-induced bonding behavior in dissimilar material welding, relevant to the development of precise and reliable microfabrication techniques.

Graphical abstract



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Keywords

Nanosecond laser; Laser welding; Dissimilar materials; Shear strength

1. Introduction

Composite structures formed by joining dissimilar materials have demonstrated significant application potential in areas such as aerospace [1], microelectronics [2], and biomedicine [3], owing to their combined advantages of lightweight construction, corrosion resistance, and excellent thermal and electrical conductivity [4]. Among these, glass-metal composites are of particular interest because they combine the advantageous properties of both materials. Conventional methods for joining glass and metal include matched sealing [5], adhesive bonding [6], brazing [7], and anodic bonding [8]. However, these techniques often encounter limitations when joining materials with large mismatches in their coefficients of thermal expansion, which can induce

thermal mismatch stresses and interfacial delamination, thereby compromising the structural integrity of the joint. In this context, laser transmission welding (LTW) has emerged as a promising alternative, offering high fluence, non-contact processing, precise control, and a minimal heat-affected zone (HAZ) [9]. These features make LTW a promising method for achieving strong, reliable bonding between transparent and metallic materials.

LTW operates by focusing a laser beam through a lens onto the substrate located beneath a transparent glass layer. This localized heating creates a molten pool that cools and solidifies into a mechanical joint [9]. Despite its advantages, continuous-wave lasers often result in thermal overloading of the glass, causing cracks and making them unsuitable for welding brittle materials [10]. As a result, researchers have turned to ultrashort pulse lasers (pulse widths <10 ps) and nanosecond pulsed lasers as alternative solutions.

Ultrashort-pulse lasers, characterized by their exceptionally high peak power and brief durations, enable strong nonlinear absorption effects that can induce localized and transient phase transitions in materials. This minimizes the HAZ and makes them particularly suitable for precision joining of brittle and transparent substrates such as glass [11]. However, the inherent limitation of their gap-bridging capability necessitates strict optical contact with nanometer-level surface precision (i.e., gap < $\lambda/4$, where λ is the laser wavelength), typically achieved through vacuum setups or mechanical clamping. Such requirements impose constraints on large-scale or flexible manufacturing. For instance, Ozeki et al. [12] achieved femtosecond laser bonding of glass and copper under fixture-applied pressure, yielding a joint strength of approximately 16MPa. Pan et al. [13] similarly realized direct bonding of sapphire to Fe-36Ni alloy using clamping, achieving strengths up to 108.35MPa. In contrast, the absence of optical contact leads to a significant decline in joint integrity. Zhang et al. [14], for example, reported a joint strength of only 2.34MPa when bonding aluminosilicate glass to various metals without any treatment of the samples or applied pressure. To address this challenge, researchers have introduced intermediate layers to enhance joint strength and permit non-optical contact welding. For example, Tian et al. [15] combined laser brazing with ultrafast laser processing to bridge micrometer-scale gaps between glass and copper using a brazing filler, achieving joint strengths up to 35MPa. Follow-up studies further confirmed the excellent thermal fatigue resistance of these joints [16]. Although these approaches have shown promise, precise control over the thickness and uniformity of intermediate layers remains challenging. Such inconsistencies often lead to non-uniform interfacial reactions and the formation of porosity defects, which compromise joint integrity. Currently, the use of external fixtures to apply pressure and maintain optical contact between workpieces continues to be the predominant approach for enhancing the stability of LTW processes. However, ultrafast laser systems are characterized by high costs, limited efficiency, and

complex maintenance requirements, all of which significantly hinder their widespread adoption in industrial manufacturing environments.

In contrast, nanosecond pulsed lasers, although generating a broader HAZ due to their reliance on linear absorption mechanisms, offer greater tolerance to interface conditions. Unlike ultrashort-pulse lasers, which require optical-grade contact and often operate in high-precision environments, nanosecond lasers can accommodate minor surface roughness and air gaps, making them suitable for practical clamping-based configurations. These advantages, coupled with higher system stability, compactness, and lower cost, make nanosecond lasers more attractive for industrial-scale joining of dissimilar materials [17,18]. Recent studies have demonstrated the feasibility of directly bonding various glass-metal pairs using nanosecond lasers, though most still rely on external mechanical clamping to maintain optical contact. For example, Feng et al. [19] achieved a shear strength of up to 33.4MPa in quartz-copper joints by applying external pressure during nanosecond laser welding. Likewise, Nguyen et al. [20,21] bonded EAGLE XG® glass to C1100 copper plates under similar pressurized conditions, reaching tensile strengths of 34.9MPa. In our previous research [22], we systematically compared clamping-assisted and ethanol-interlayer assisted techniques for preserving optical contact, highlighting their effect on joint morphology and mechanical integrity. Although promising results have been reported, a fundamental understanding of how key laser parameters govern interfacial morphology, thermal behavior, and defect evolution remains limited. Consequently, the process window for nanosecond LTW in dissimilar material systems remains largely undefined.

In this study, a nanosecond pulsed LTW approach is employed to achieve direct optical contact bonding between 304 stainless steel and quartz glass using mechanical clamping. This work aims to elucidate the laser-induced interfacial dynamics, including localized energy deposition, absorption mechanisms, and microstructural evolution at the bonding zone. By systematically adjusting laser power, frequency, scanning speed, and scanning repetitions, the study explores their impact on interfacial morphology, bonding characteristics, and fracture behavior. These findings provide valuable insights into laser-matter interaction and interface evolution, supporting the advancement of high-precision LTW techniques for reliable integration of dissimilar materials in optical and microengineering applications. While recent studies have validated the feasibility of nanosecond LTW under pressurized conditions, a comprehensive parametric investigation coupled with quantitative interfacial morphology classification and mechanical correlation remains absent. This study addresses this gap by establishing a direct link between process parameters, morphological zones (smooth/fragment/ablation), and fracture behavior. Such systematic mapping contributes to the deeper understanding and optimization of nanosecond LTW processes for dissimilar materials. Overall, this

work distinguishes itself by bridging practical nanosecond laser welding with in-depth interfacial analysis, offering new pathways for industrial applications of LTW without relying on ultrashort-pulse systems or advanced surface preparation.

2. Experimental details

2.1. Materials and surface preparation

In this study, quartz glass ($\phi 20\text{ mm} \times 2\text{ mm}$, 99.9% purity, red-light transmittance of 99%) and 304 stainless steel ($35\text{ mm} \times 35\text{ mm} \times 0.7\text{ mm}$) were selected as the base materials for the welding experiments.

Quartz glass plates were used as received (commercial grade) and only ultrasonically cleaned in anhydrous ethanol prior to welding. No additional mechanical polishing was applied. The baseline surface topography of the as-received glass was characterized using a 3D digital microscope (VHX-6000, Keyence), yielding an average apparent roughness of $R_a \approx 0.60\mu\text{m}$ (Fig. S1(a)). This value is instrument- and scale-dependent and is reported here to describe the micrometer-scale surface waviness/defects of the commercial-grade glass, rather than the nanoscale roughness of optically polished glass. For optically polished glass with nm-level roughness, AFM is typically required for accurate nanoscale topography quantification. Similar AFM-based characterization of pristine glass has been reported in the literature [23,24]. The 304 stainless-steel substrates were commercially polished (as received). Prior to welding, they were only ultrasonically cleaned in anhydrous ethanol and dried. The surface roughness of the as-received stainless steel was measured by a VHX-6000 3D optical microscope, giving an average $R_a \approx 1.25\mu\text{m}$ (Fig. S1(b)). Using a finished smooth surface helps provide a stable and repeatable interfacial contact condition under clamping during LTW.

The thermophysical properties of the materials and the chemical composition of 304 stainless steel are listed in Table 1, Table 2, respectively. The thermophysical properties listed in Table 1 were obtained from standard materials databases and technical datasheets [19,25]. The chemical composition of 304 stainless steel is provided in Table 2, based on the nominal ranges defined by the ASTM A240 standard [26]. In addition, to clarify the chemical purity of the glass, trace impurity elements were analyzed and are listed in Table 3, as detected by the professional quality inspection center.

Table 1. The thermophysical properties of 304 stainless steel and quartz glass.

Material	T _{Melt} (°C)	σ _{Tensile} (MPa)	Coefficient of linear expansion	Thermal diffusivity (mm ² /s)	Thermal conductivity (W/(m·K))
304 Stainless steel	1440	520	17.2×10 ⁻⁶	3.835	14.986
Quartz glass	1723	40-70	5.5×10 ⁻⁷	1.3	0.95

Table 2. The weight percentage (wt%) of various components in 304 stainless steel.

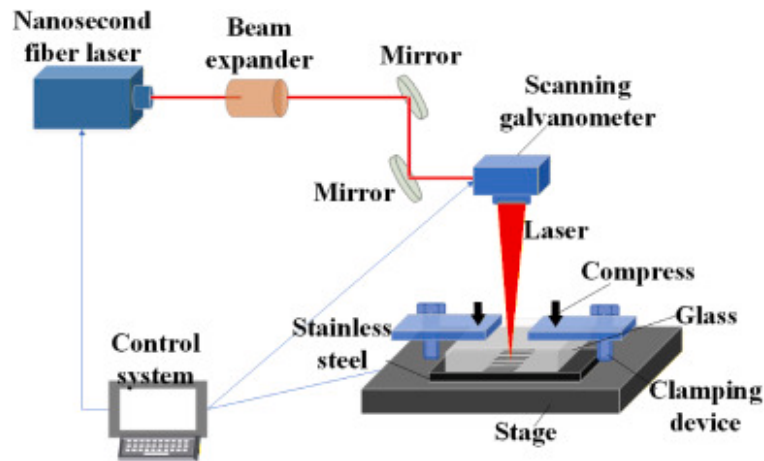
Component	C	Mn	Si	P	S	Ni	Cr	N	Fe
wt (%)	0.049	0.93	0.44	0.028	0.004	8.00	18.03	0.062	72.457

Table 3. Trace impurity composition of the quartz glass used in this study.

Element	Al	Ca	Fe	K	Mg	Na	Ti
Concentration (mg/kg)	8.55	0.86	0.83	0.27	0.04	1.96	5.57

2.2. Laser welding system and optics

The experimental setup for the laser welding system is illustrated in Fig. 1. The system consists of a nanosecond fiber laser (YDFLP-E-60-M7-M-R, JPT, Shenzhen), a beam expander, reflective mirrors, and a scanning galvanometer. The laser operates at a wavelength of 1064nm, with a maximum output power of 60W. The pulse duration is fixed at 200 ns, and the focal spot diameter is approximately 50μm. This value represents the focal spot at the lens focus in air.



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Fig. 1. Schematic diagram of the laser welding system.

Considering the transmission of the laser beam through a 2mm-thick quartz glass plate (refractive index $n=1.46$), the focal shift was calculated based on the Gaussian beam propagation and refraction at planar interfaces, as discussed in Refs. [[27], [28], [29]]. The effective focal length shift Δf can be estimated by:

$$\Delta f = d \cdot \left(1 - \frac{1}{n}\right) \quad (1)$$

where d is the quartz thickness and n is its refractive index. Substituting the values into the formula, the result indicates that the laser focal point is shifted forward (toward the quartz-steel interface) by approximately 0.63mm compared to its nominal focal length in air. The impact of this focal shift was considered during process optimization, and the actual welding focal position was adjusted accordingly.

Although the laser pulse duration is specified, the exact temporal pulse shape was not recorded. Depending on the laser source, nanosecond pulses may exhibit non-uniform profiles with leading-edge intensity spikes, which could influence the absorption and energy coupling efficiency. This aspect will be considered in future work to more accurately model the interaction dynamics.

To ensure optical contact between the glass and stainless steel specimens during welding, external pressure was applied using a custom-designed clamping fixture consisting of gaskets and screws. The laser is focused on the surface of the stainless steel, and the welding process is executed by controlling the scanning path via a computer interface.

2.3. Welding procedure and process design

The specific LTW process is shown in Fig. 2. As depicted in Fig. 2(a), the stainless steel and quartz glass samples were used as received (steel: commercially polished; glass: commercial grade) and were ultrasonically cleaned in anhydrous ethanol for 5 min and dried with absorbent paper prior to welding. A mechanical clamping fixture was subsequently used to apply external pressure, minimizing the interfacial gap and fixing the relative positions of the materials to ensure optical contact. The applied pressure was measured in real time using a flexible thin-film sensor (RunesKee) and recorded via a computer interface, as shown in Fig. 2(b). During welding, the applied force was maintained at approximately 65 N. Subsequently, the laser beam was transmitted through the glass and focused onto the surface of the stainless steel substrate to initiate welding. The weld was executed using a horizontal line-array scanning pattern. As illustrated in Fig. 2(c), the scanning area was fixed at 2.5 mm × 2.5 mm, comprising uniformly spaced parallel lines with a line pitch of 25 μm.



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Fig. 2. (a) Schematic diagram of the LTW process. (b) Schematic structure of the flexible thin-film sensor placed between the glass and stainless steel plate. (c) Scanning path schematic for the laser beam.

To investigate the effects of laser power, scanning speed, frequency, and scanning times on weld strength, four groups of single-factor controlled experiments were conducted. In each group, one parameter was varied across 5 to 7 discrete levels, while all

other parameters were held constant. The specific laser processing parameters are summarized in Table 4. The parameter ranges were determined based on preliminary tests and literature data, with focus on regions where significant variations in bonding quality were anticipated. During each trial, when varying one parameter, the other three were held at mid-range values within the preliminary optimal region to maintain stable process conditions.

Table 4. Laser processing parameters used in nanosecond laser transmission welding experiments.

Parameters	Value
Power (W)	6.6, 7.8, 9, 10.2, 11.4, 12.6, 13.8
Scanning speed (mm/s)	40, 60, 80, 100, 120, 140, 160
Frequency (kHz)	20, 30, 40, 50, 60, 70, 80
Scanning times (times)	10, 30, 50, 70, 90

Since both scanning speed and frequency influence the overlap ratio of the laser spots, the corresponding calculation formula is provided in equation (2) to facilitate subsequent analysis [30].

$$S_{overlap} = \varphi = \frac{\Delta x}{\Delta y} = 1 - \frac{v_{scan}}{f_{rep} \times d_f} \quad (2)$$

In equation (2), $S_{overlap}$ denotes the spot overlap ratio; v_{scan} is the scanning speed (mm/s); d_f represents the focused spot diameter (μm); and f_{rep} is the laser pulse frequency (kHz). Δx is the center-to-center spacing between consecutive laser pulses along the scan direction, and Δy is the distance between adjacent scan lines in the transverse direction.

In addition to geometric parameters, the single-pulse laser fluence F (J/cm^2) was also considered to evaluate the energy input per unit area by a single laser pulse. It was calculated according to the following expression [31]:

$$F = \frac{P}{f \cdot A} \quad (3)$$

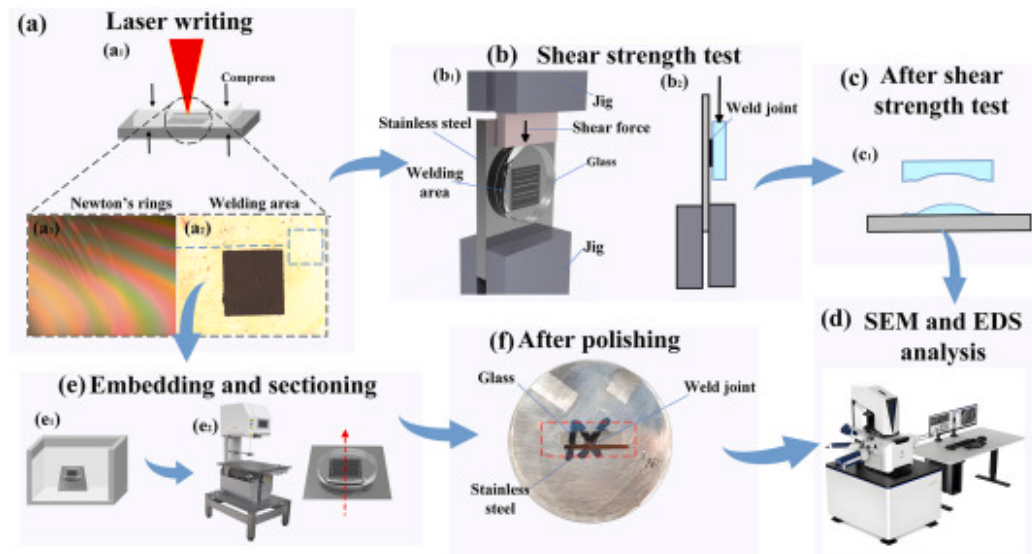
Where P is the average laser power (W), f is the pulse repetition frequency (Hz), and A is the effective laser spot area (cm^2), calculated as $A = \pi(d/2)^2$, with d being the focused spot diameter.

2.4. Characterizations

2.4.1. Shear strength test

The shear strength of the welded joints was measured using a digital tensile tester (AILIGU, Shenzhen) to evaluate the welding quality. A schematic of the test setup is shown in Fig. 3(b), specifically subfigures (b₁-b₂). During testing, the welded workpiece was fixed onto the lower fixture such that the direction of the applied shear force was precisely perpendicular to the weld path. A pressing head, aligned parallel to the edge of the glass, applied a downward force to the glass surface. The critical force at which separation occurred between the stainless steel and the glass was recorded. It is important to ensure that the point of force application is vertically aligned with the center of the welded region. This alignment avoids the introduction of bending moments, which could otherwise interfere with the accuracy of the experimental results. The shear strength (σ) was calculated using equation (4) [32].

$$\sigma = \frac{F}{S} \quad (4)$$



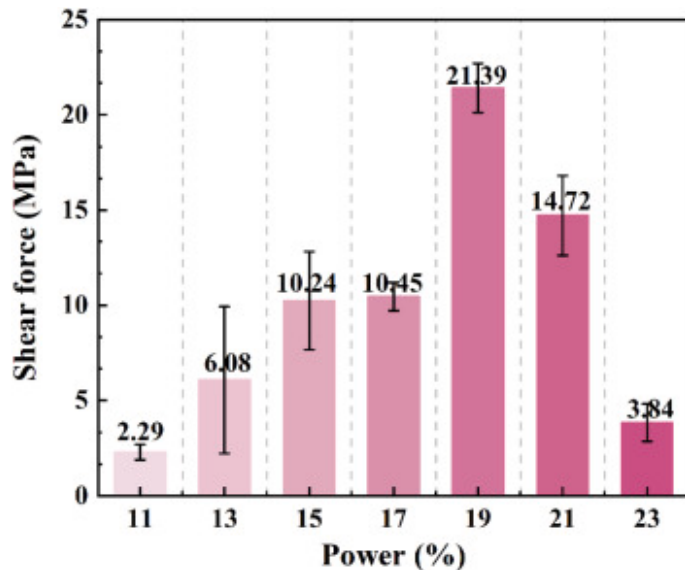
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Fig. 3. Process flow diagrams for fracture surface and cross-section characterization testing. (a-d) represent the fracture surface testing sequence; (a, e, f, d) represent the cross-sectional testing sequence.

In equation (4), F denotes the applied shear force, and S represents the welded area.

For each set of laser parameters, three replicate specimens were tested. The reported values represent the mean, and the error bars in Fig. 4, Fig. 9, Fig. 12, Fig. 15 indicate the standard deviation across the three samples.



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Fig. 4. Effect of laser power on the shear strength of welded joints. The frequency (40kHz), scanning speed (100mm/s), and number of scans (30) were kept constant during the experiments. Error bars indicate standard deviation ($n = 3$).

2.4.2. Fracture surface characterization

The fracture surface characterization analysis procedure is illustrated in Fig. 3(a-d). The term fracture surface refers to the interface between the quartz glass and 304 stainless steel in the welded region after shear strength testing, as exemplified in Fig. 3(c). The complete characterization process is described as follows: First, LTW was performed, as shown in Fig. 3(a). An optical

microscope (UMM-IF-5) was employed for preliminary inspection of the weld zone morphology, as depicted in Fig. 3(a₂-a₃). Distinct Newton's rings were observed on the surface, confirming that the samples remained in optical contact after welding. Subsequently, the welded composite structures underwent shear strength testing to induce separation, as shown in Fig. 3(b and c). The resulting fracture surface, presented in Fig. 3(c), was examined in detail using a digital stereomicroscope (VHX-6000, Keyence) to observe the macroscopic two-dimensional (2D) and three-dimensional (3D) topographies of the contact interface. The VHX-based 3D profilometry is used to characterize the micrometer-scale morphology of the welded/fractured regions. It does not provide reliable quantitative roughness for nm-scale pristine-glass topography, for which AFM is typically required [23,24]. Finally, as illustrated in Fig. 3(d), a field emission scanning electron microscope (FE-SEM, ZEISS Sigma 360) was used to analyze the micro/nanoscale morphology and structural features of the fracture surface. Elemental composition analysis was carried out using an energy-dispersive X-ray spectroscopy (EDS) system (OXFORD Xplore 30). High-resolution X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha, America) was utilized to elucidate the chemical bonding mechanisms at the welded interface.

2.4.3. Cross-section characterization

The procedure for cross-section characterization analysis is illustrated in Fig. 3(a)–(e), (f), and (d). The cross-section refers to a cut made along the weld joint of the welded composite structure to expose the internal bonding interface. As shown in Fig. 3(e), the welded workpiece was first cold-mounted using an embedding resin. The mounted samples were then sectioned along the weld line using a diamond wire-cutting machine (Metkon, Micracut 201), allowing for complete exposure of the weld region. Following sectioning, one side of the exposed cross-section was polished to obtain a smooth surface for observation, as illustrated in Fig. 3(f). Finally, as shown in Fig. 3(d), SEM was employed to observe the microstructural features of the weld cross-section. Moreover, EDS was used to perform both area and line scans to analyze the elemental distribution across the interface.

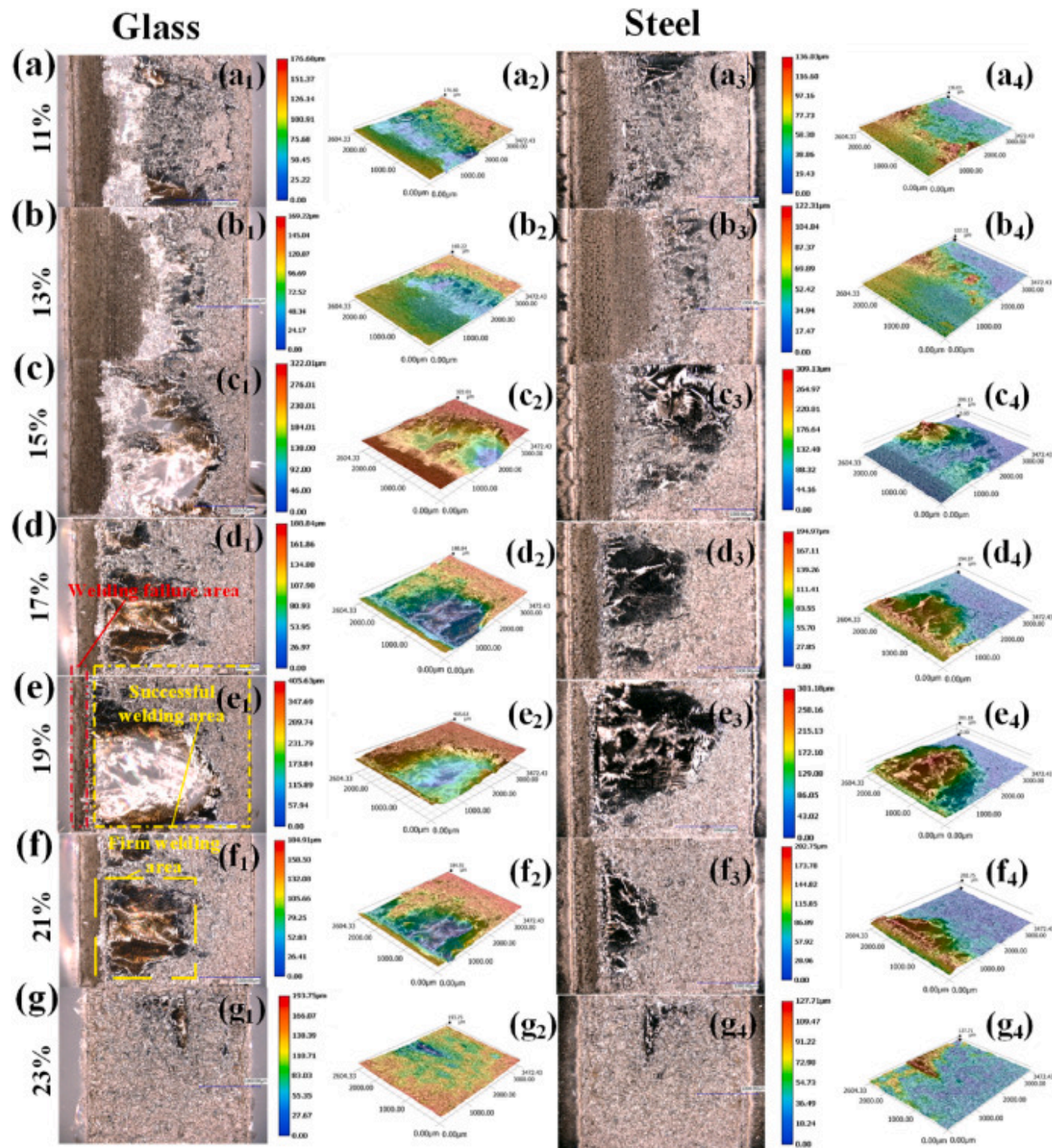
3. Results and discussions

3.1. Influence of process parameters on joint strength and morphology

3.1.1. Effect of laser power

With the experimental setup and characterization methods established, we now present the results regarding how varying laser parameters affect joint strength, morphology, and interface quality. In this group of single-variable experiments, laser power was varied from 6.6W to 13.8W, while laser frequency (40kHz), scanning speed (100mm/s), and number of scans (30) were kept constant. One scan refers to a complete coverage of the programmed horizontal line matrix across the sample surface. [Fig. 4](#) illustrates the effect of laser power on the shear strength of the welded joints. As the laser power increased, the corresponding single-pulse energy also rose. The shear strength exhibited a trend of initial enhancement followed by a subsequent decline. At relatively low power levels (6.6 W-7.8W), although bonding between glass and stainless steel could be achieved, the resulting shear strength remained limited. As the power increased further (9W-12.6W), the joint strength improved significantly, reaching a peak value of 21.39MPa at 11.4W power. However, when the power was further raised to 13.8W, a noticeable decrease in shear strength was observed. These results demonstrate that laser power plays a critical role in determining joint integrity. An appropriate power range (9W-12.6W) facilitates sufficient energy input and stable molten pool formation, thereby enhancing weld strength. In contrast, excessively low power may result in inadequate bonding, whereas excessive power can introduce thermal damage or interfacial stress, ultimately deteriorating joint quality [\[33\]](#).

[Fig. 5](#) presents the fracture surface morphologies of the welded joints under various laser power levels, observed using a digital stereomicroscope. Specifically, images (a₁)-(g₁) display the 2D fracture surfaces on the glass side, while (a₂)-(g₂) show the corresponding 3D profiles. Similarly, (a₃)-(g₃) illustrate the 2D fracture surfaces on the stainless steel side, and (a₄)-(g₄) present the corresponding 3D images. In the 3D representations, the horizontal and vertical axes denote the width and depth of the fracture surface, respectively, while the color scale indicates height variation across the plane. The presence of pronounced undulations on the originally smooth bonding interface indicates successful bonding between the stainless steel and the quartz glass [\[18,34\]](#). Furthermore, the shear strength can be indirectly assessed by observing the size of depressions on the glass side or the extent of raised features on the stainless steel side (the amount of glass lost from the glass surface or the quantity of glass adhered to the steel substrate). For instance, the yellow-marked region in [Fig. 5\(f₁\)](#) represents an area of firm welding. Comparing the dimensions and elevation of such firm welding zones under different power levels reveals that the largest and highest welding area occurs at 11.4W laser power, with a peak height reaching 301.18μm, corresponding to the maximum shear strength of 21.39MPa reported in [Fig. 4](#).



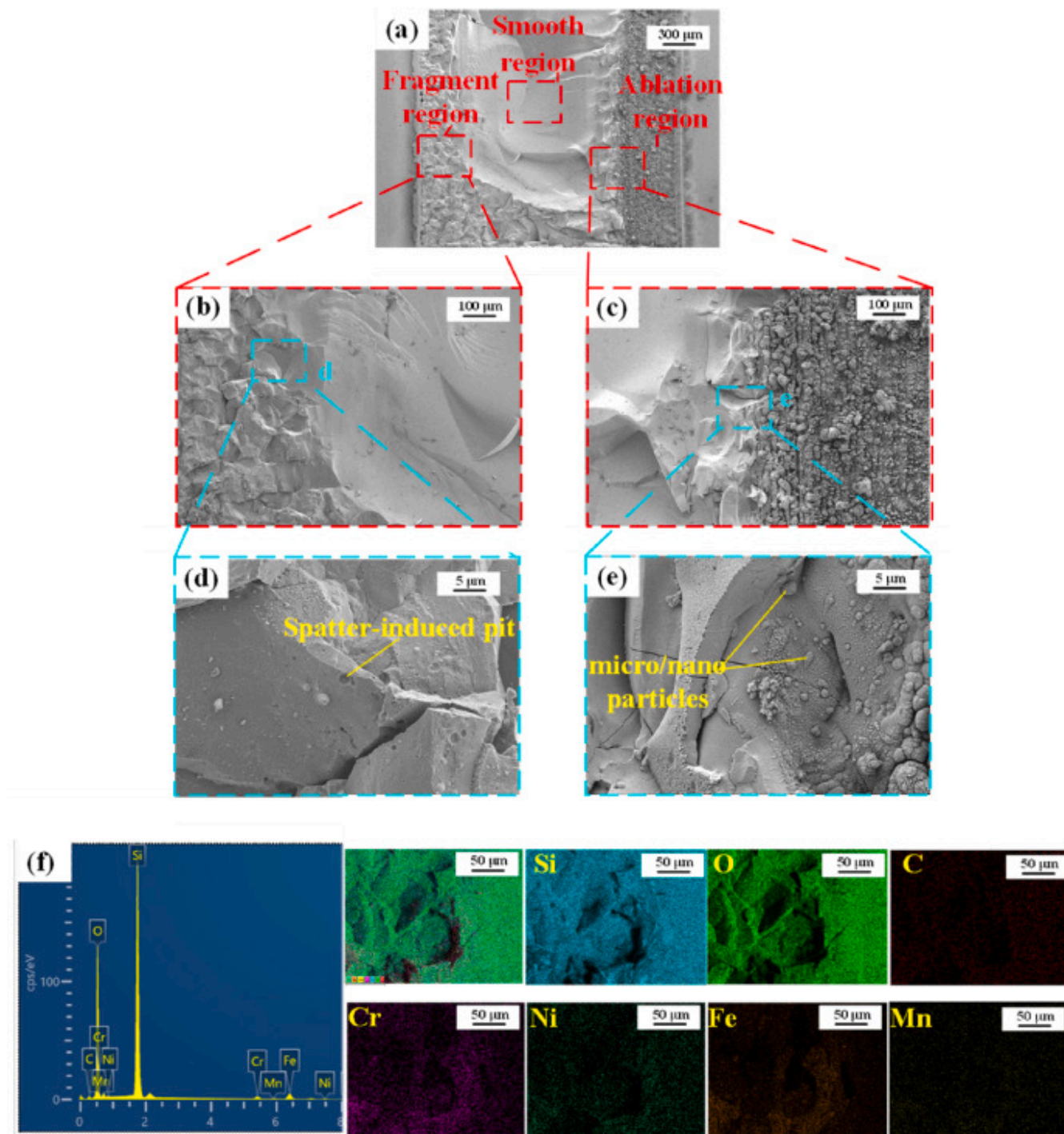
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Fig. 5. Fracture surface morphologies under different laser power levels: (a₁)-(g₁) show 2D images of the glass-side fracture surfaces; (a₂)-(g₂) show the corresponding 3D topographies of the glass side; (a₃)-(g₃) display 2D images of the stainless steel-side fracture surfaces; and (a₄)-(g₄) present the corresponding 3D topographies of the stainless steel side.

As shown in the 2D images of Fig. 5(a₁)-(g₁) and Fig. 5(a₃)-(g₃), the laser-irradiated area can be categorized into two distinct zones: (1) the successful welding area, such as the yellow box in Fig. 5(e₁), where a resolidified molten pool forms a visible bond, and glass residues are clearly observed on the stainless steel side after fracture; and (2) the welding failure area, such as the red box in Fig. 5(e₁), where no joint is formed. In these regions, only linear ablation marks from the laser scan are observed, and the surface remains smooth with no glass adhesion after separation. As the laser power increases, the boundary between the successful welding area and the welding failure area progressively shifts, resulting in an expansion of the bonded region and a reduction in the unbonded region. However, when the power exceeds the optimal threshold of 12.6W, as shown in Fig. 5(g₁)-(g₄), the successful welding area is nearly absent. Significant glass ablation debris appears at the interface, indicating thermal overstress. This excessive energy input induces micro-cracks and localized perforations. Most of the glass material disintegrates into particles without forming a molten pool or bonding with the stainless steel, thereby severely degrading the joint strength and structural reliability [22,35].

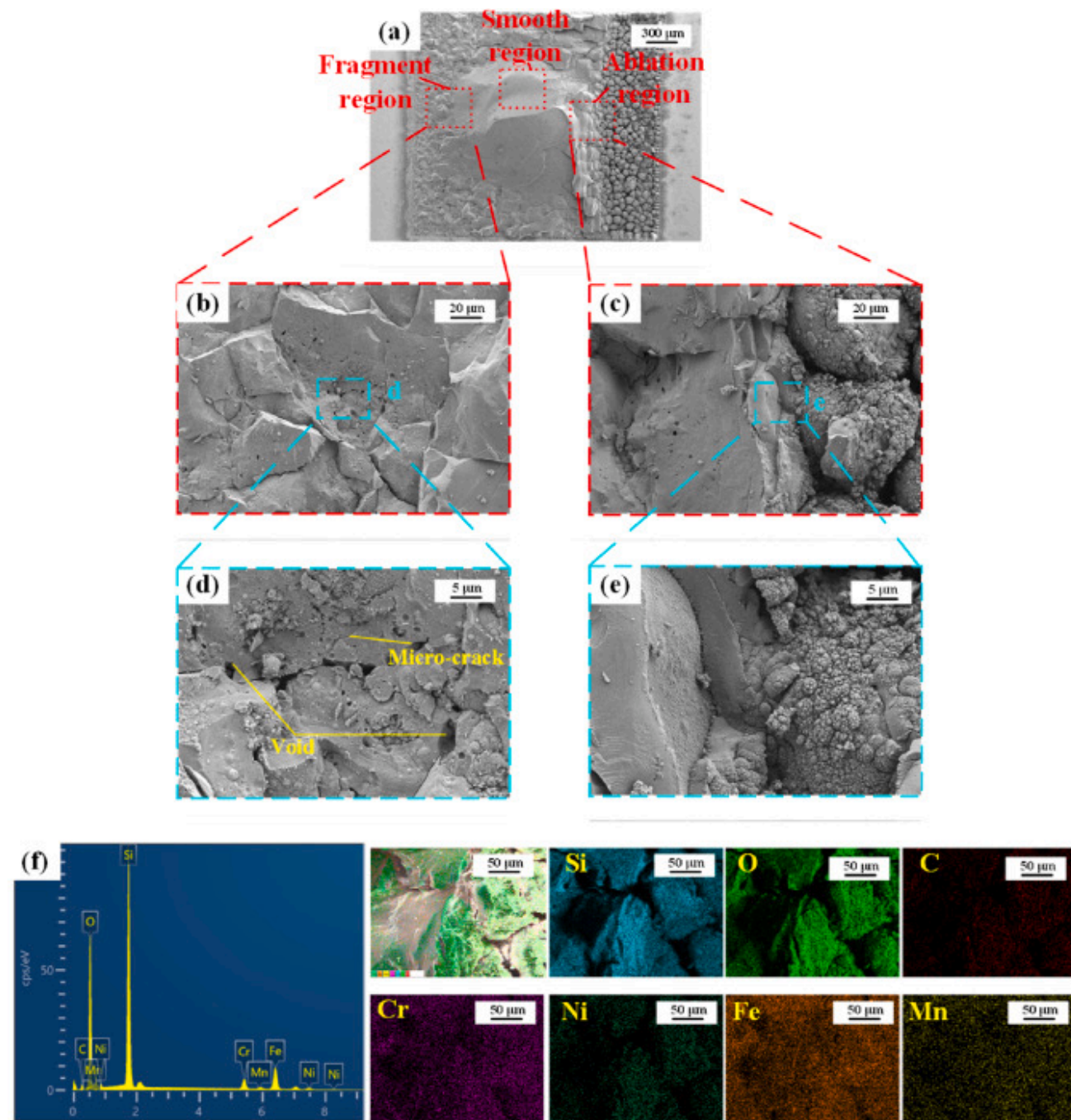
SEM and EDS mapping were conducted on the fracture surface after shear testing. The sample selected for analysis was welded under the conditions of 11.4W laser power, 40kHz frequency, 100mm/s scanning speed, and 30 scanning times, corresponding to a maximum shear strength of 21 MPa, as shown in Fig. 6, Fig. 7.



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Fig. 6. SEM and EDS mapping results of the glass-side fracture surface: (a)–(e) SEM images of the glass side; (f) elemental distribution map of the fracture surface obtained by EDS mapping.



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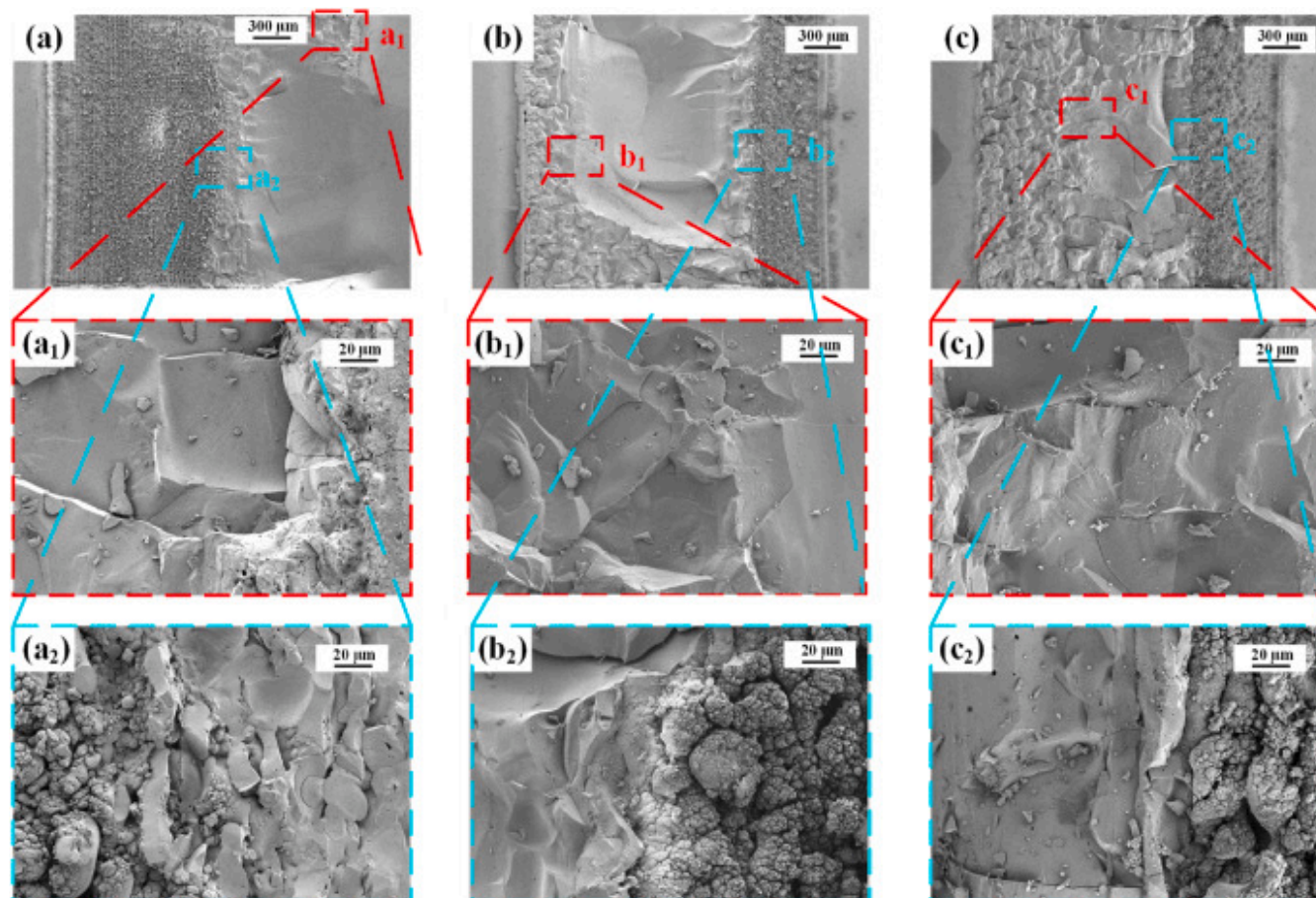
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Fig. 7. SEM and EDS analysis results of the stainless steel-side fracture surface: (a)–(e) SEM images of the stainless steel side; (f) elemental distribution map obtained by EDS mapping.

Fig. 6 shows the glass-side fracture surface morphology. As seen in Fig. 6(a), the processed region can be divided into three distinct zones, consistent with the earlier 2D and 3D morphological analyses. First, the central area exhibits a relatively smooth surface, referred to as the smooth region. The magnified view of this zone is shown on the right side of Fig. 6(b). This area corresponds to the firm welding area identified in Fig. 5. It features a concave pit formed by cohesive fracture within the glass during shear testing, where large fragments of glass were torn from the bulk and transferred to the stainless steel side. Owing to its internal origin and crystalline smoothness, this region indicates strong bonding and high shear strength. Second, the fragment region, located on the left side of Fig. 6(a) and magnified in Fig. 6(b), is characterized by numerous raised and sunken fragments. This area is produced when fracture occurs near the focal zone on the glass side. Although bonding is achieved, the resulting joint is mechanically weaker. The glass that adheres to the steel surface in this zone is scattered and fragmented, with visible debris on the fracture surface. This region corresponds to the successful welding area in Fig. 5, excluding the firm bonding region. Third, the rightmost part of the fracture surface, referred to as the ablation region, shows clear linear traces of laser ablation and corresponds to the welding failure area in Fig. 5. As shown in the magnified view in Fig. 6(c), no molten pool was formed in this area during welding. Instead, regular groove structures were produced by direct laser ablation of the material. Further magnification of Fig. 6(c), shown in Fig. 6(e), reveals numerous submicron and nanoscale particles distributed along the edges of the ablation grooves. These features are attributed to laser-material interactions, where localized heating, vaporization, and ionization occur, followed by ablation and plasma-induced deposition, ultimately leading to the formation of fine particulate structures on the surface [36]. In addition, Fig. 6(d) reveals several small and regularly shaped circular pits and protrusions distributed across the fracture surface. These microstructures are attributed to the splashing of molten glass and metal material from the weld pool during laser processing. Upon solidification, the splashed material becomes embedded within the weld seam, and following fracture, corresponding pit and protrusion features appear on the glass and stainless steel sides, respectively. This interpretation is further supported by the EDS surface mapping of the fragment region shown in Fig. 6(f), where Fe, Cr, and Ni, which were originally absent in the glass, are found to be uniformly distributed across the glass surface. These findings indicate that material mixing and elemental diffusion occurred between the glass and stainless steel at the joint interface.

Fig. 7 presents the SEM and EDS analysis results of the stainless steel-side fracture surface after the shear test. As shown in Fig. 7(a), the surface can be divided into three distinct regions, referred to as the smooth region, the fragment region, and the ablation region. These zones correspond to those previously identified on the glass side in Fig. 6(a). Fig. 7(b) and (c) display magnified views of selected areas from Fig. 7(a), where Fig. 7(b) highlights the fragment region, and the right portion of Fig. 7(c) shows the ablation region. In the fragment region illustrated in Fig. 7(b), numerous brittle fracture features are observed. These are characterized by irregular concave and convex structures formed by fractured glass fragments adhered to the stainless steel surface. A higher magnification in Fig. 7(d) reveals the presence of voids and micro-cracks. These findings indicate that the fragment region underwent molten pool formation and solidification during the welding process. The micro-cracks are attributed to excessive thermal stress generated during the cooling phase due to significant laser energy accumulation on the quartz glass. The formation of cavities and voids is explained by the violent interaction between the laser and materials, during which absorbed laser energy generates a mixed molten pool accompanied by high-temperature vapor and gas bubbles at the interface. Rapid cooling after laser irradiation traps these bubbles before they can collapse or escape, leading to the formation of cavities and voids in the weld seam [37]. The existence of these micro-defects significantly reduces the shear strength of the welded joint by inhibiting the formation of a compact and continuous joint. Furthermore, numerous micro-cracks were observed along the weld seam in the region near the laser focus on the glass side. During the shear test, these pre-existing micro-cracks act as fracture initiation sites, leading to separation along these weakened paths. As a consequence, a firmly bonded area similar to the smooth region is not formed. Instead, the interface is predominantly composed of brittle fracture zones corresponding to the fragment region. Fig. 7(f) shows the EDS elemental mapping of the fragment region on the stainless steel side. The presence of Si, a key component of glass, distributed across the stainless steel surface confirms that material mixing and elemental diffusion occurred during laser welding. The resulting interfacial composition indicates the formation of a mixed molten pool and supports the establishment of a bonded joint between glass and stainless steel.

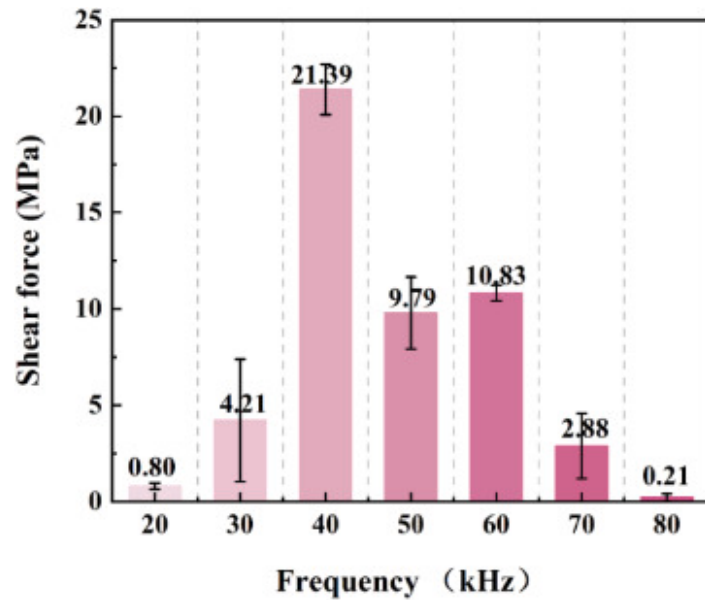
SEM analyses of the fracture surfaces under different laser power conditions were conducted, and the results for the glass-side fracture surfaces are shown in Fig. 8. Consistent with the earlier morphological classification, the fracture surfaces at various power levels can still be divided into three primary regions: the smooth region, the fragment region, and the ablation region.



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Fig. 8. SEM images of the glass-side fracture surfaces under different laser powers: (a) fracture surface at 7.8W laser power; (b) fracture surface at 11.4W laser power; (c) fracture surface at 13.8W laser power.



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Fig. 9. Effect of frequency on the shear strength of the welded joints. The laser power (11.4W), scanning speed (100mm/s), and number of scans (30) were kept constant.

Fig. 8(a) shows the SEM image of the fracture surface obtained under a laser power of 7.8W. The enlarged views of the fragment region and ablation region in Fig. 8(a) are presented in Fig. 8(a₁) and Fig. 8(a₂), respectively. It can be clearly observed that the ablation region occupies a large proportion of the processed area, nearly half of the total region, indicating a high fraction of welding failure. This is attributed to insufficient laser energy absorption at the interface between the materials, which prevents the formation of a molten pool in certain zones. As a result, direct bonding between the glass and stainless steel fails to occur, leading to a significant reduction in joint strength. The fracture occurs near the focal region, corresponding to a lower mechanical performance. Fig. 8(c) displays the fracture surface under a high laser power of 13.8W. The magnified views of the fragment and ablation regions are shown in Fig. 8(c₁) and Fig. 8(c₂), respectively. Although the increase in laser power suppresses the extent of the ablation region, the area occupied by the fragment region becomes notably larger, while the smooth region that contributes to firm bonding becomes significantly smaller. Compared with the SEM image at moderate power in Fig. 8(b), the fragment region under high power displays more frequent and densely distributed brittle fractures. The excessive heat input at high laser

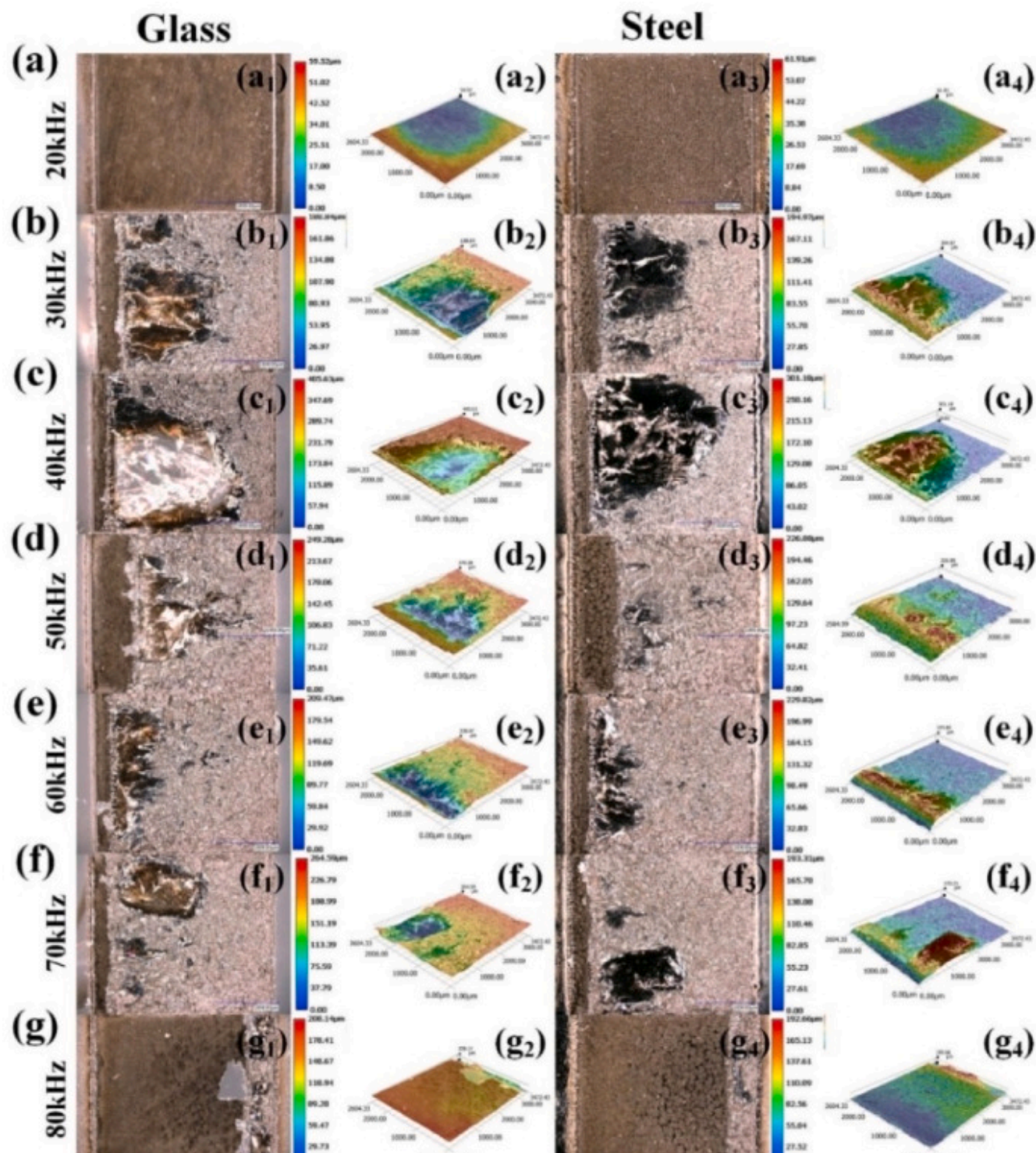
power results in substantial thermal accumulation on the glass side, inducing high thermal stress during cooling. This thermal stress leads to the formation of numerous micro-cracks and irregular voids, thereby compromising the mechanical strength of the welded joint [38].

At low power levels, the laser energy is insufficient to fully melt the two materials, resulting in poor shear strength. Additionally, a larger ablation region often appears under low power, reducing the effective welding area and significantly impairing joint performance. In contrast, at moderate power levels, the increased energy input allows more material to melt and mix, which subsequently solidifies into a larger fusion zone. This promotes stronger interfacial bonding and enhances mechanical properties. However, further increases in power cause excessive melting and vaporization of material, enlarging the HAZ and intensifying thermal stress at the interface. The resulting turbulent melt flow and splashing introduce additional defects, such as micro-cracks and voids. Consequently, the weld fails along the crack paths near the focal region, forming large fragmented areas with poor bonding strength. Although it is generally expected that ablation regions decrease with increasing laser power due to improved melting, Fig. 8(c) shows an expanded ablation area at 13.8W. This is attributed to over-ablation resulting from excessive thermal accumulation and vaporization pressure, which caused localized spallation and material ejection rather than effective bonding. Therefore, the relationship between ablation region and laser power is non-monotonic, showing initial suppression of ablation followed by re-expansion under overheating conditions.

3.1.2. Effect of frequency

Following the influence of laser power, we now examine how laser pulse frequency contributes to the joint formation and interfacial bonding characteristics. Fig. 9 shows the effect of laser frequency on the shear strength of welded joints. During this set of experiments, the laser power (11.4W), scanning speed (100mm/s), and number of scans (30) were held constant. As the laser frequency increased from 20kHz to 80kHz, the shear strength initially improved and then declined, indicating a non-monotonic dependence. According to the spot overlap rate defined in equation (2), a higher pulse frequency results in greater spot overlap. In pulsed laser welding, a weld seam is formed by a series of overlapping micro-molten pools. As the laser beam moves, the heat from a previous pulse may not fully dissipate before the next pulse arrives, leading to cumulative thermal effects. This phenomenon becomes significant when the time interval between successive laser pulses is shorter than the time required for the absorbed heat to diffuse out of the focal volume [39]. At low frequencies (e.g., 20kHz), the shear strength remains relatively low. In the intermediate frequency range (40-60kHz), the shear strength improves significantly. However, when the frequency is further increased to 80kHz, the shear strength of the joints drops sharply, approaching zero.

The 2D and 3D morphologies of the welded joints under different frequencies are shown in Fig. 10. Similar to the analysis of laser power variations in Fig. 5, the area and height of the firm welding region are maximized at an optimal frequency of 40kHz, where the stainless steel side exhibits the greatest amount of adhered glass. This corresponds to the highest measured shear strength of the joint, as illustrated in Fig. 10(c₁) through Fig. 10(c₄). At lower frequencies, such as 20kHz, linear ablation marks appear directly on both sides of the weld due to insufficient energy input. The entire surface of the material remains uniform, and no glass adhesion is observed in the processed zone. The welding failure area occupies nearly the entire region, aligning with the low shear strength observed under this condition, as shown in Fig. 10(a₁) through Fig. 10(a₄). Conversely, when the frequency is excessively high (e.g., 80kHz), extensive over-ablation occurs. The successful welding area becomes very limited, resulting in poor mechanical performance of the joint, as shown in Fig. 10(g₁) through Fig. 10(g₄).



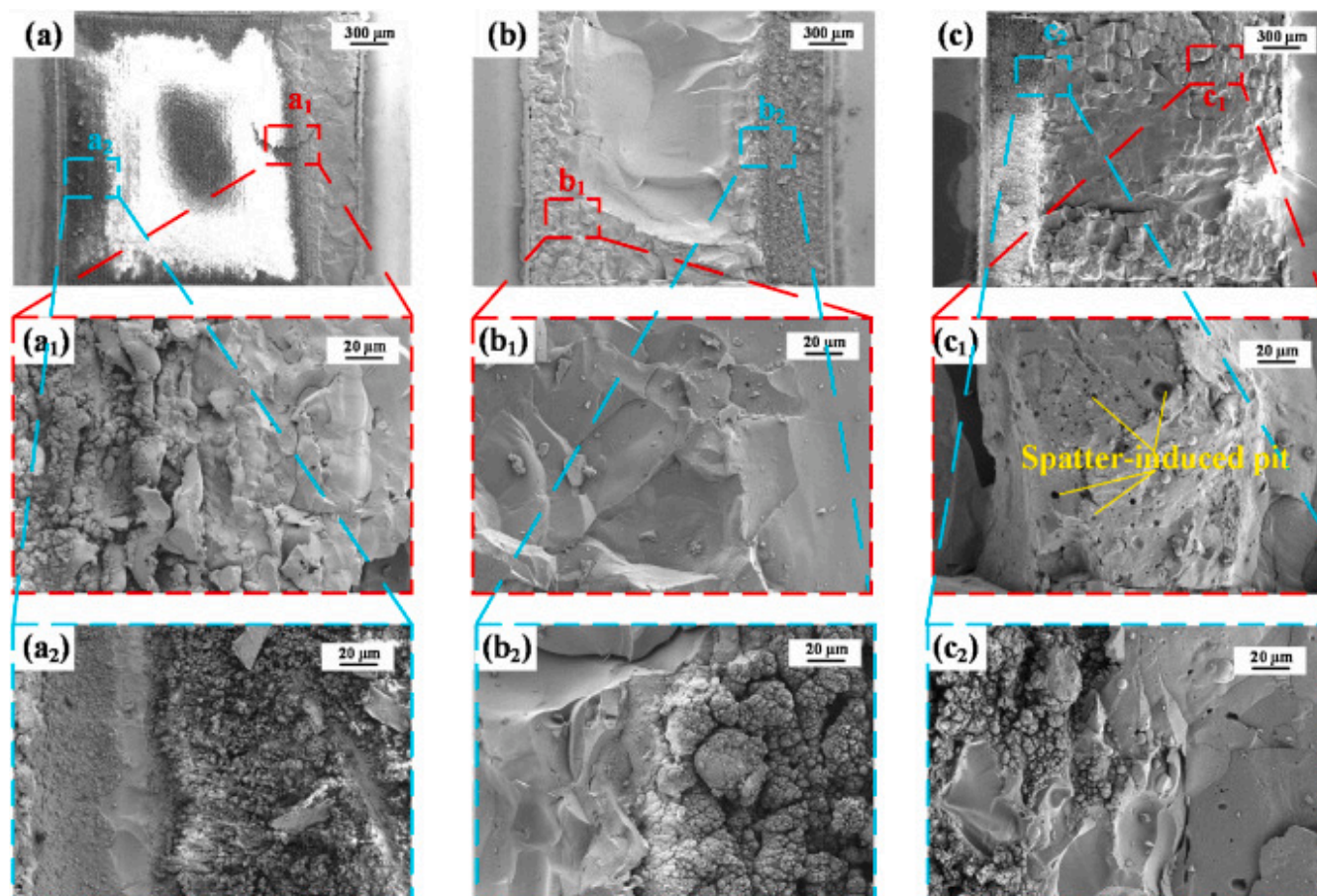


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Fig. 10. (a)–(g) Fracture surface morphologies at different frequencies: (a₁)–(g₁) 2D images of the glass-side fracture surfaces; (a₂)–(g₂) 3D images of the glass-side fracture surfaces; (a₃)–(g₃) 2D images of the stainless steel-side fracture surfaces; (a₄)–(g₄) 3D images of the stainless steel-side fracture surfaces.

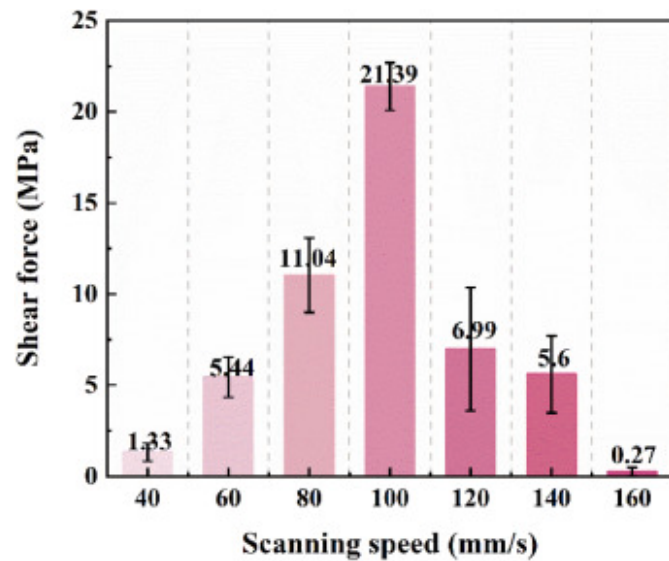
The SEM images of fracture surfaces at different laser pulse frequencies are presented in Fig. 11. Fig. 11(a) shows the fracture surface under a laser frequency of 30kHz, with Fig. 11(a₁) and Fig. 11(a₂) displaying magnified views of the fragment region and ablation region, respectively. A prominent white area is observed at the center of Fig. 11(a), which can be attributed to the charging effect induced by the electron beam during SEM observation. Since glass has poor electrical conductivity, charge accumulates on its surface, generating localized electrostatic fields that appear as bright regions in the SEM image [40]. This phenomenon also indirectly indicates insufficient mixing between the glass and stainless steel, with a relatively low metallic content on the glass side. Fig. 11(c) displays the fracture surface at a laser frequency of 70kHz, where a large proportion of the region is dominated by brittle fracture features, indicating a wide distribution of fragment regions. Fig. 11(c₁), a magnified view of the fragment region in Fig. 11(c), reveals the presence of numerous cavities and micro-cracks. In comparison, the corresponding region at 40kHz shown in Fig. 11(b₁) exhibits fewer such defects. These microstructural defects significantly compromise joint strength, explaining the sharp decrease in shear performance at higher frequencies. Fig. 11(c₂), a close-up of the ablation region, also reveals extensive particle ejection caused by over-ablation.



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Fig. 11. SEM images of the glass-side fracture surfaces at different frequencies: (a) fracture surface at 30kHz; (b) fracture surface at 40kHz; (c) fracture surface at 70kHz.



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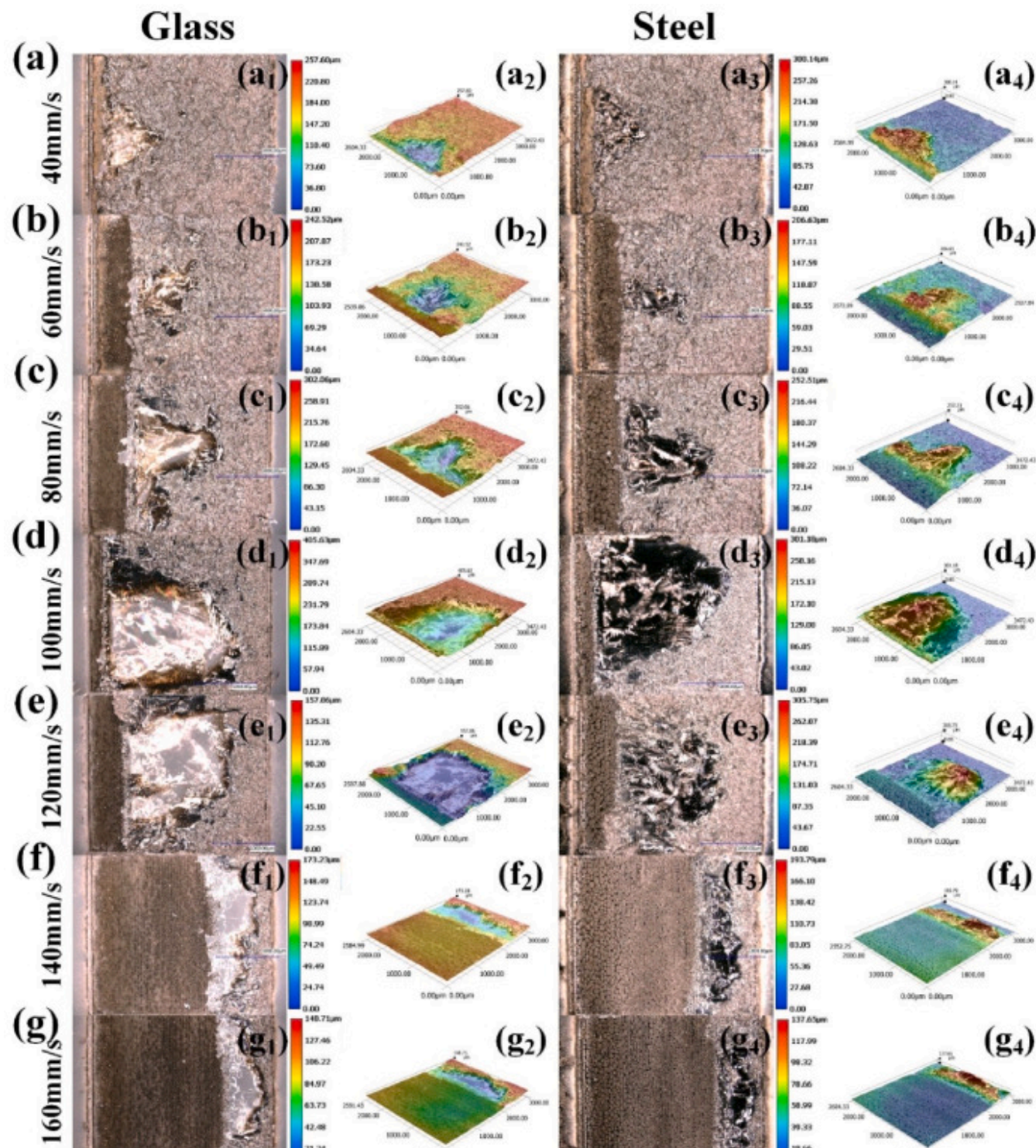
Fig. 12. Effect of scanning speed on the shear strength of the welded joint. The laser power (11.4W), frequency (40kHz), and number of scans (30) were kept constant.

At lower frequencies, the thermal accumulation effect is weak, and the energy is insufficient to fully melt the two materials, resulting in mere surface ablation rather than the formation of a molten pool. This leads to weak bonding strength. At intermediate frequencies, the energy input becomes more uniform and sufficient, promoting better mixing and fusion at the interface, thereby creating a more extensive bonded region with enhanced joint integrity upon solidification. In contrast, excessively high frequencies lead to strong thermal accumulation and high spot overlap rates, causing severe ablation of the stainless steel surface, particle ejection, and widespread formation of fragment regions. Furthermore, excessive frequencies can destabilize the molten pool through fluid turbulence and splashing, thereby undermining the weld quality and mechanical strength.

3.1.3. Effect of scanning speed

Next, we investigate the effect of scanning speed on the joint quality. The effect of scanning speed on the shear strength of the welded joints is illustrated in Fig. 12. In this group of experiments, the laser power (11.4W), frequency (40kHz), and number of scans (30) were held constant. A distinct nonlinear relationship is observed between scanning speed and joint strength. As the scanning speed increases, the shear strength initially rises, reaches a peak, and then declines. At relatively low scanning speeds (e.g., 40mm/s), the joints exhibit lower shear strength. When the scanning speed is in the moderate range (80-100mm/s), the shear strength improves significantly. However, further increases in scanning speed (e.g., 160mm/s) lead to a noticeable reduction in shear strength.

Fig. 13 illustrates the 2D and 3D morphologies of the fracture surfaces at various laser scanning speeds. By analyzing the separated interfaces on both the glass and stainless steel sides, a clear correlation can be observed between scanning speed, molten pool characteristics, and joint quality. As shown in Fig. 13(a₁)-(a₄), at a low scanning speed of 40mm/s, although the area of the successfully welded region is relatively large, the size and depth of the firm welding area remain limited. This indicates that while material mixing occurs, the molten pool is shallow and insufficiently extended to form a robust bonding zone, resulting in relatively low shear strength. As the scanning speed increases to the intermediate range of 80-100mm/s, the welded area becomes significantly larger, and the molten pool exhibits clear and uniform contours, as shown in Fig. 13(c) and (d). The 3D profiles reveal a synchronized increase in pit depth on the glass side and protrusion height on the stainless steel side. Notably, at a scanning speed of 100mm/s, the maximum height of the firmly bonded region reaches 301.10μm, corresponding to the peak shear strength of 21.39MPa reported in Fig. 12. This suggests that moderate scanning speeds promote enhanced molten pool stability and enable effective wetting at the interface, resulting in dense and reliable joints upon solidification. However, when the scanning speed is excessively high (e.g., 160mm/s), the welded region appears discontinuous. As shown in the 3D morphology, numerous unclosed fusion boundaries emerge on the glass surface. Under such conditions, both the area and height of the firm welding area sharply decrease, with the maximum protrusion height reduced to only 137.65μm, accompanied by a dramatic drop in joint strength.



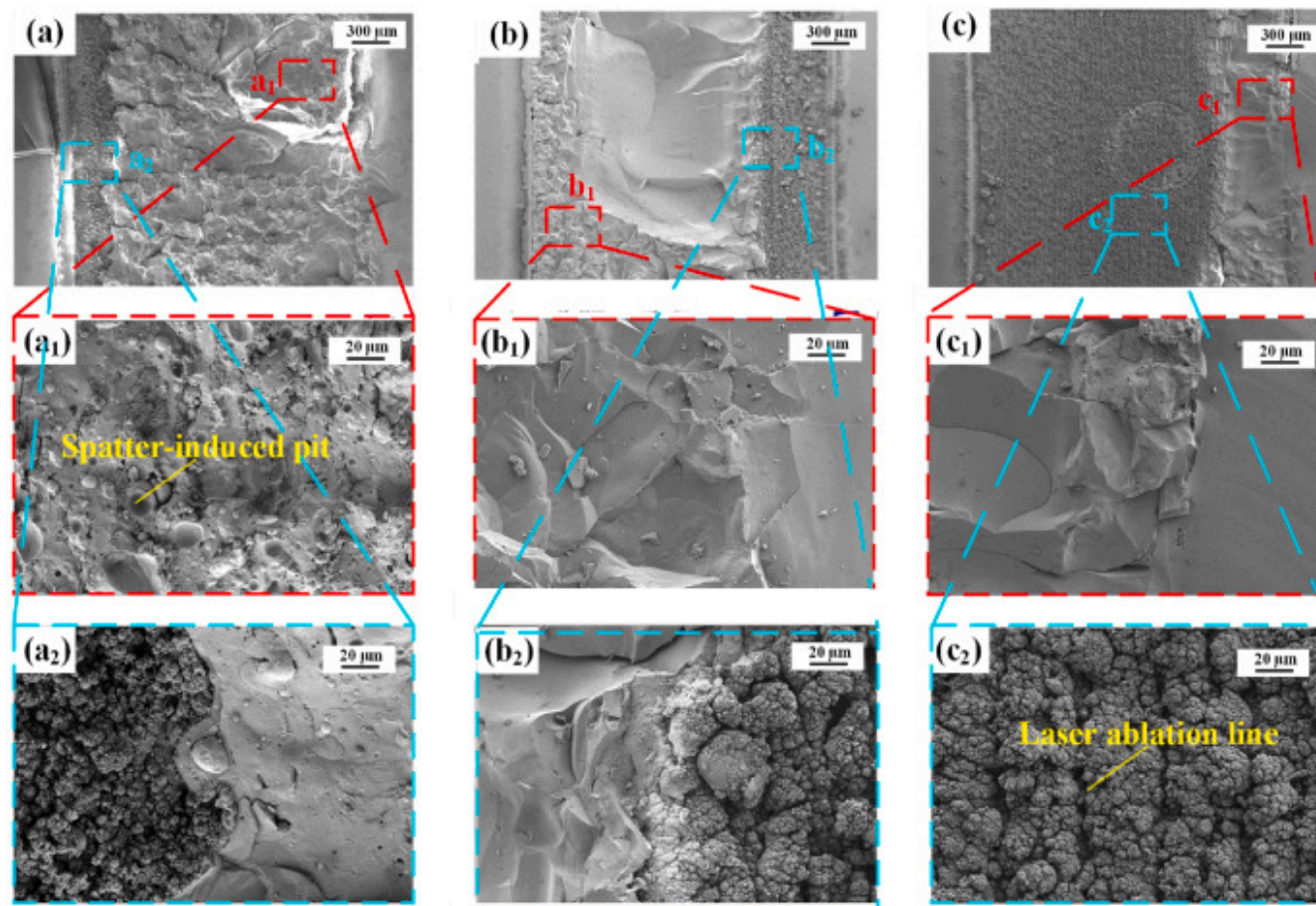


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Fig. 13. Fracture surface morphologies under different laser scanning speeds: (a₁)-(g₁) 2D images of the glass-side fracture surfaces; (a₂)-(g₂) 3D images of the glass-side fracture surfaces; (a₃)-(g₃) 2D images of the stainless steel-side fracture surfaces; (a₄)-(g₄) 3D images of the stainless steel-side fracture surfaces.

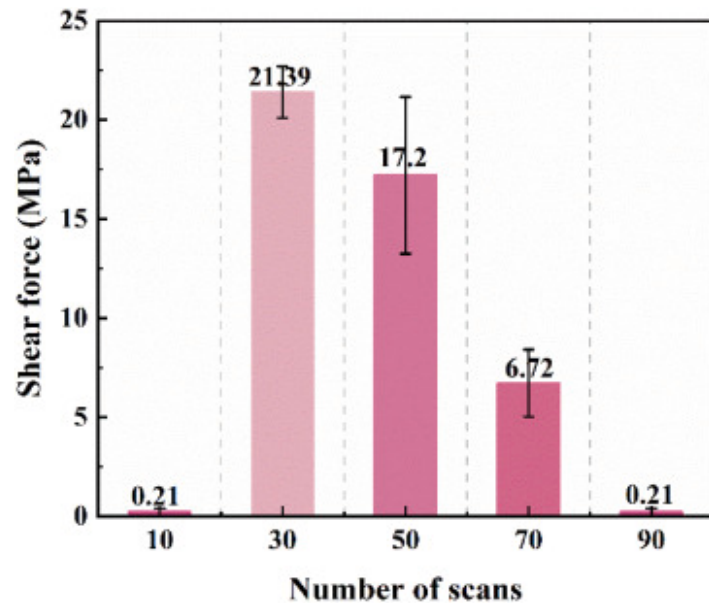
The SEM images of fracture surfaces under different laser scanning speeds are presented in Fig. 14. Fig. 14(a) shows the fracture surface obtained at a scanning speed of 40mm/s, while Fig. 14(a₁) and 14(a₂) are magnified views of the fragment region and ablation region, respectively. Compared to the fracture surface at 100mm/s, the fragment region at 10mm/s is densely populated with spatter-induced pits, indicating extensive melt pool activity and the presence of significant spatter within the molten zone. Fig. 14(c) illustrates the fracture surface at a scanning speed of 160mm/s, with Fig. 14(c₁) and 14(c₂) depicting the enlarged fragment and ablation regions. Due to the high scanning speed and reduced spot overlap, insufficient material mixing occurred, resulting in a large ablation region where clear laser ablation lines can be observed. This condition leads to a reduction in joint strength.



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Fig. 14. SEM images of glass-side fracture surfaces at different scanning speeds: (a) fracture surface at 40mm/s; (b) fracture surface at 100mm/s; (c) fracture surface at 140mm/s.



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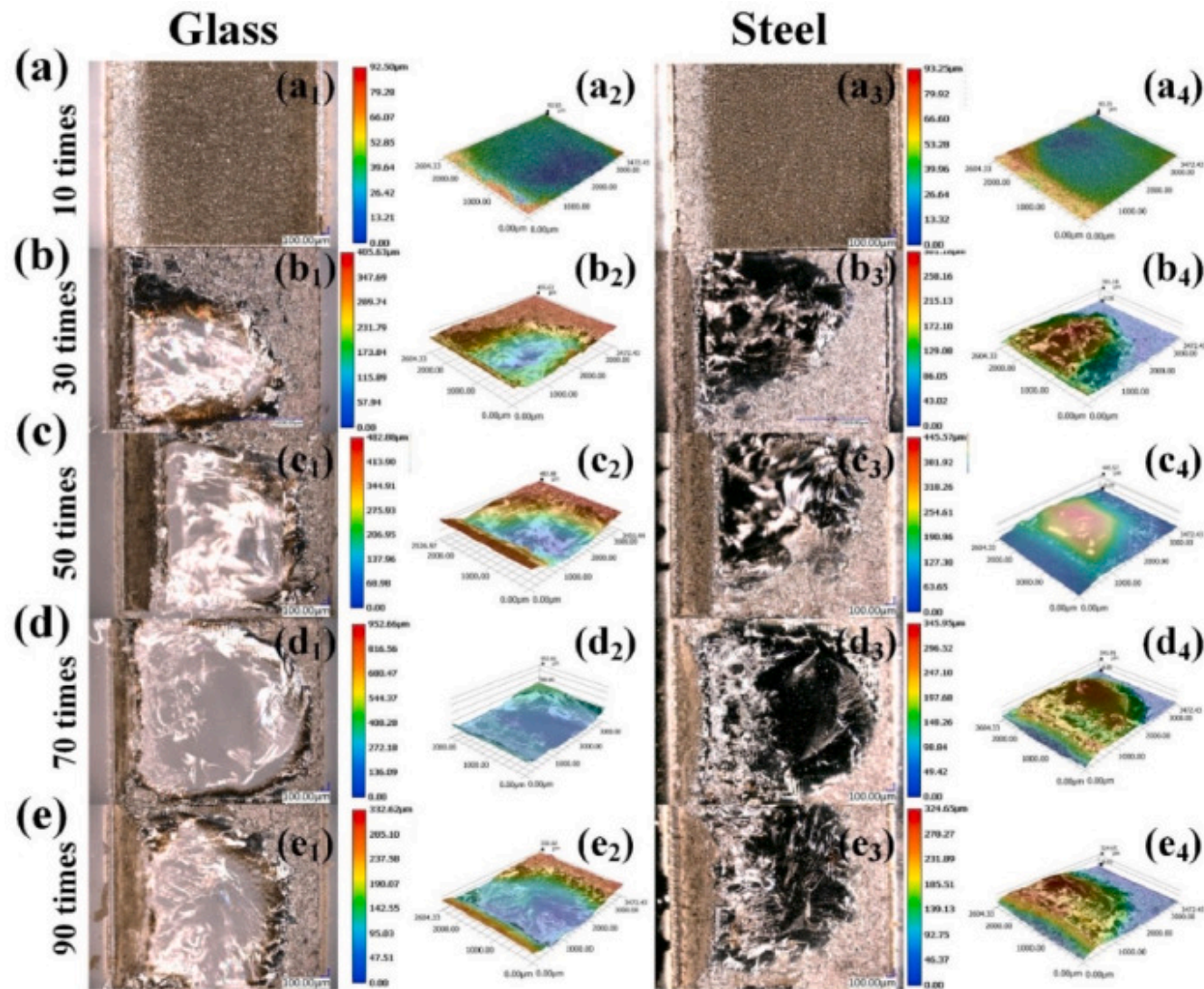
Fig. 15. Effect of scanning times on the shear strength of the joints. The laser power (11.4W), frequency (40kHz), and scanning speed (100mm/s) were held constant.

According to equation (2), a lower scanning speed increases the spot overlap ratio and enhances thermal accumulation. Under constant processing parameters, excessive spot overlap may cause surplus laser energy absorption, leading to substantial thermal accumulation and concentrated thermal stress. Upon cooling, the melt pool solidifies rapidly, forming brittle phases at the focal interface and inducing defects such as spatter and thermally-induced micro-cracks, ultimately compromising the joint strength [41]. In contrast, moderate scanning speeds offer balanced heat input, facilitating melt pool formation and material mixing while minimizing thermal stress and associated defects, thereby improving mechanical performance. At excessively high scanning speeds, the reduced overlap and insufficient thermal accumulation prevent effective melt pool development. Furthermore, the increased laser recoil pressure primarily ablates material rather than promoting bonding, resulting in diminished joint strength and the presence of laser ablation lines on the fracture surface [42].

3.1.4. Effect of scanning times

Subsequently, we evaluate how multiple scan passes influence the bonding behavior. The effect of scanning times on the shear strength of the joints is illustrated in Fig. 15. In this group of experiments, the laser power (11.4W), frequency (40kHz), and scanning speed (100mm/s) were held constant. It is observed that the shear strength increases initially and then decreases with increasing scanning times. At a low number of scanning times (10), the joints exhibit relatively low shear strength. As the scanning times increase to a moderate range (30-50 times), the shear strength improves significantly. However, when the scanning times further increase (up to 90 times), a notable decline in shear strength is observed.

The 2D and 3D morphologies of the fracture surfaces under different scanning times are shown in Fig. 16. As seen in Fig. 16(a₁)-(a₄), when the scanning times are relatively low (10 times), a large welding failure area is observed in the 2D images of the glass surface, and the successful welding area occupies only a small fraction. The interface shows clear delamination, and the 3D morphology indicates insufficient protrusion height on the stainless steel side, suggesting only partial material fusion. With insufficient thermal accumulation, the surface fails to receive adequate energy input to facilitate effective material mixing, resulting in a minimal molten pool. In contrast, as shown in Fig. 16(b₁)-(b₄) and Fig. 16(c₁)-(c₄), when the scanning times increase to 30-50, the molten pool becomes more continuous, and the depth of pits on the glass side and the height of protrusions on the stainless steel side significantly increase in the 3D images. The proportion of firmly bonded regions also increases, corresponding to the peak shear strength of 21.39MPa. This indicates that moderate thermal accumulation promotes the formation of a uniform fusion layer. However, as illustrated in Fig. 16(e₁)-(e₄), when the scanning times are further increased to 90, the amount of glass adhering to the stainless steel does not decrease significantly, yet the shear strength drops sharply. Excessive thermal input leads to substantial thermal accumulation, resulting in large temperature gradients upon rapid cooling. This creates excessive localized thermal stress, inducing brittle fracture and the formation of numerous micro-cracks in the glass, which dramatically reduces joint strength [42].

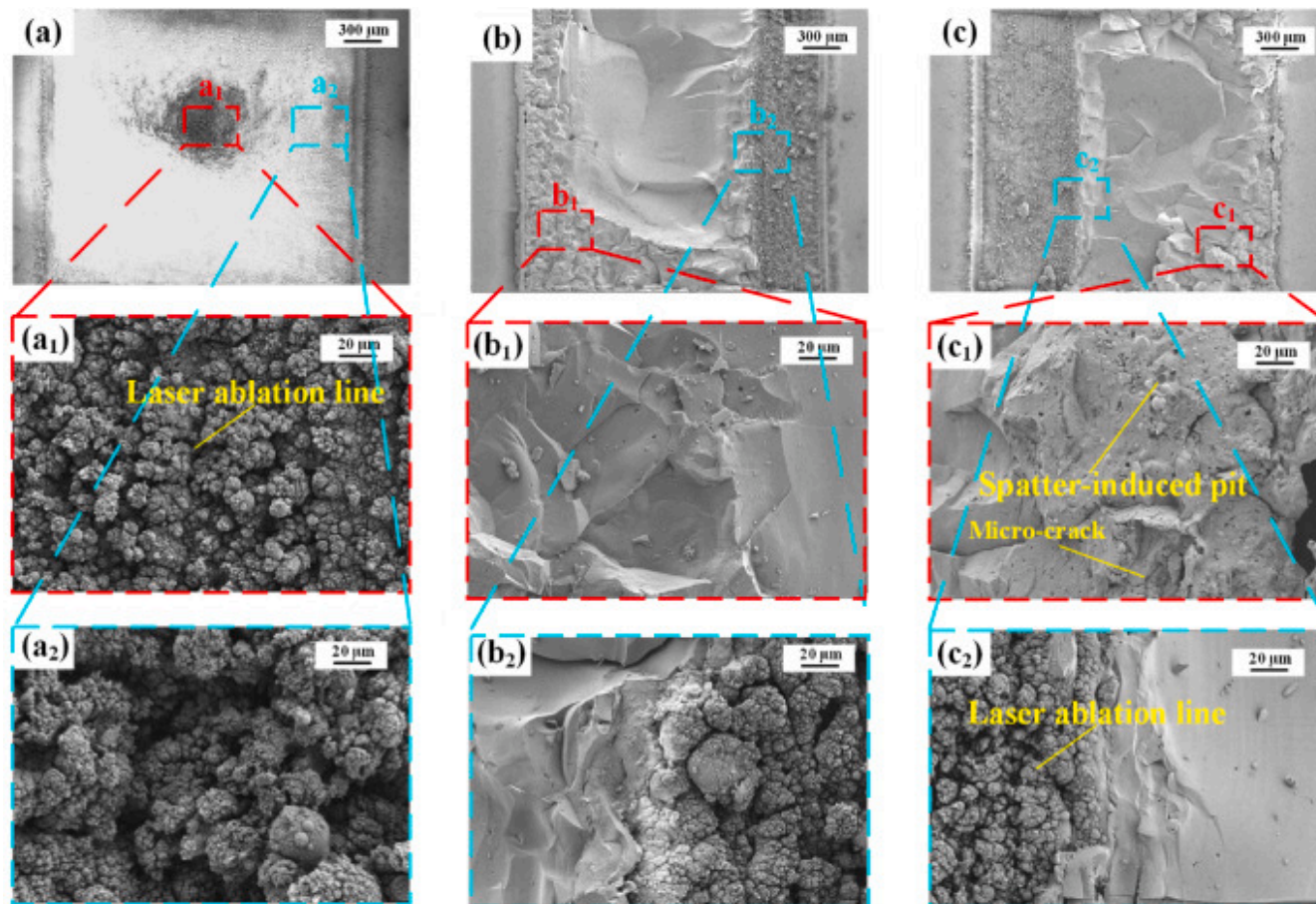


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Fig. 16. (a)–(e) Fracture surface morphologies at different scanning times: (a₁)–(e₁) 2D images of the glass-side fracture surfaces; (a₂)–(e₂) 3D images of the glass-side fracture surfaces; (a₃)–(e₃) 2D images of the stainless steel-side fracture surfaces; (a₄)–(e₄) 3D images of the stainless steel-side fracture surfaces.

The SEM images of fracture surfaces at different scanning times are shown in Fig. 17. Fig. 17(a) presents the fracture surface obtained with 10 scanning times. Fig. 17(a₁) and 17(a₂) are enlarged views of the ablation region marked in Fig. 17(a). Similar to the observations in Fig. 11(a), white high-brightness areas appear due to charging effects, which are caused by the poor electrical conductivity of the glass. This indicates that at a low number of scans, the laser primarily induces surface ablation on the glass side without sufficient melting or material mixing. As a result, no direct joint is formed between the two materials in this region. Fig. 17(c) displays the fracture surface at 70 scanning times. Fig. 17(c₁) and 17(c₂) show magnified views of the fragment region and ablation region, respectively. In Fig. 17(c₁), a large number of spatter-induced pits and micro-cracks can be observed, which contrasts sharply with the relatively uniform and defect-free morphology seen at 30 scanning times in Fig. 17(b₁). Furthermore, deep grooves are visible in Fig. 17(c₂), attributed to repeated laser ablation at the same location.



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Fig. 17. SEM images of fracture surfaces at different scanning times: (a) fracture surface at 10 scanning times; (b) fracture surface at 30 scanning times; (c) fracture surface at 70 scanning times.

When other processing parameters are held constant, a low number of scanning times leads to insufficient heat accumulation on the material surface, preventing full melting and wetting of the glass and stainless steel. This results in the formation of extensive ablation regions, compromising the shear strength of the joint. At moderate scanning times, heat accumulation is more balanced, effectively promoting the formation of the molten pool, improving material mixing, reducing the HAZ, and minimizing thermal stress and defects. However, excessive scanning times cause excessive heat input and accumulation, leading to stress concentration and the formation of numerous micro-cracks within the thermally expanded glass. During the shear test, the joint tends to fracture along these internal micro-cracks near the defocused region, thereby significantly reducing the shear strength.

To sum up, based on the laser parameters investigated in this section (power: 6.6-13.8W; frequency: 20-80kHz; spot diameter: 50 μ m), the single-pulse laser fluence was calculated using equation (3). The resulting fluence values ranged from 8.4J/cm² to 17.6J/cm² as power increased, and from 29.0J/cm² to 7.25J/cm² as frequency increased, under fixed spot diameter conditions. Since scanning speed and number of scans do not influence the energy per pulse, they do not affect the fluence directly. By correlating the calculated fluence values with the measured shear strength and fracture surface morphologies, three distinct energy input regimes were identified. When the fluence was below approximately 13.0J/cm², insufficient energy input resulted in poor bonding or ablation-dominated regions, as evidenced by the minimal glass transfer. In the intermediate fluence range of approximately 13.0-16.0J/cm², the joints exhibited optimal bonding, as indicated by peak shear strength values and fracture morphologies characterized by deep pits, cohesive fractures, and considerable material mixing at the interface. When the fluence exceeded 16.0J/cm², excessive thermal input caused interfacial defects such as microcracks, irregular voids and large fragmented regions, thus leading to reduced mechanical performance. These observations collectively suggest that the effective fluence window for robust bonding between quartz glass and stainless steel lies within the range of 13.0-16.0J/cm².

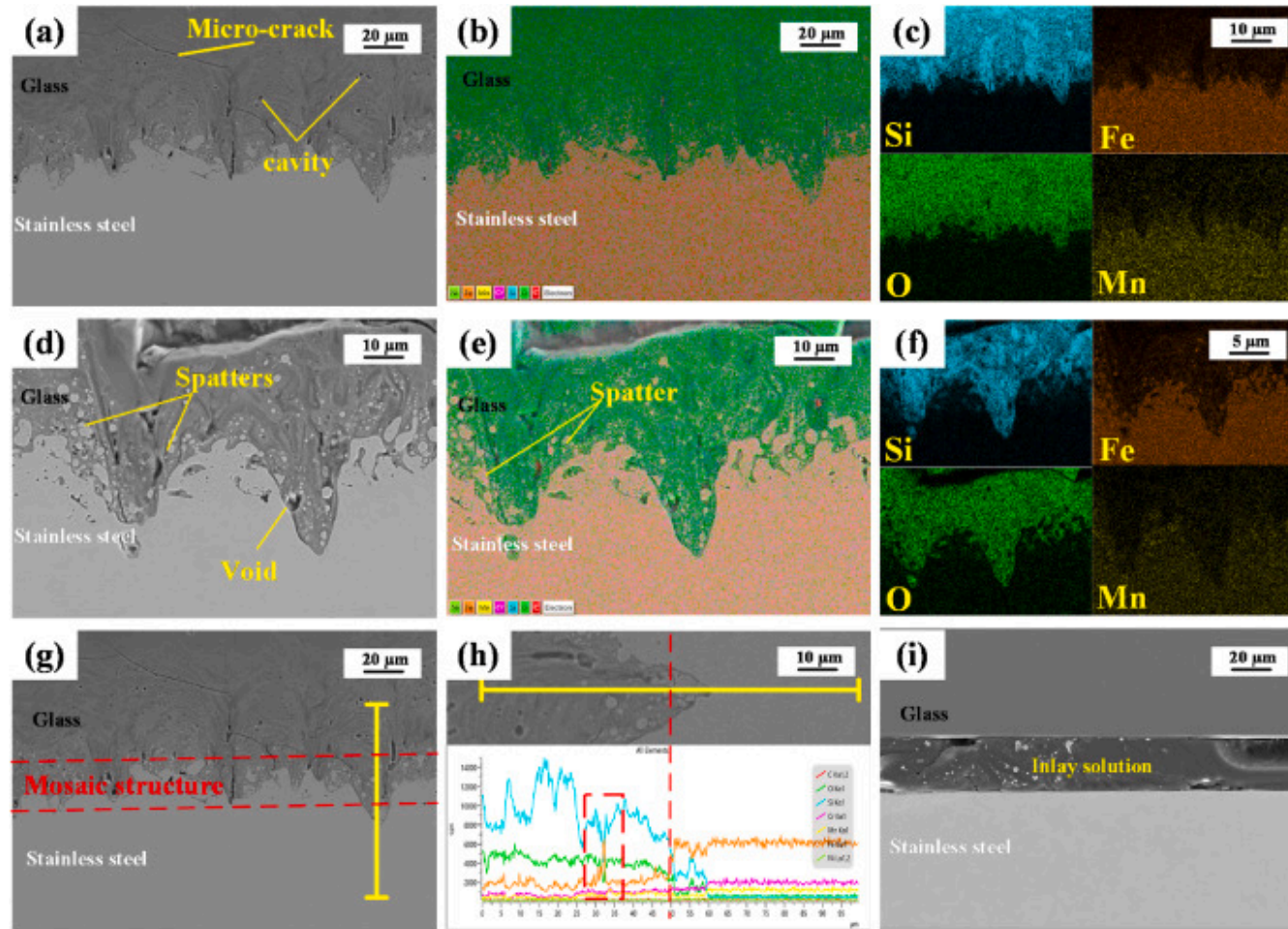
3.2. Microstructural analysis of weld cross-sections

To further investigate the interfacial bonding mechanism, the cross-section of a welded sample obtained under optimized processing parameters was examined. The sample was fabricated using 11.4W laser power, 40kHz pulse frequency, 100mm/s

scanning speed, and 30 scanning times, which corresponded to the maximum shear strength (21.39MPa) achieved in this study, as detailed in Section 3.1.

Although this specimen exhibited the best mechanical performance among all tested conditions, it was selected for cross-sectional analysis not solely due to its superior strength, but rather because it displayed a clearly identifiable weld seam and representative microstructural features. In particular, it revealed typical characteristics of the bonding interface, including molten pool morphology, transition layers, and interfacial defects, which were consistently observed across other well-bonded samples. Therefore, this specimen provides a suitable and representative basis for analyzing the interfacial structure and failure mechanisms in quartz-glass-to-stainless-steel laser welding.

Fig. 18(a) shows the SEM image of the cross section of the weld seam. Fig. 18(b) and (c) present the corresponding EDS elemental mapping results for the area in Fig. 18(a). It can be observed that the stainless steel and glass are tightly bonded at the weld interface, forming a distinct mosaic structure. The EDS maps reveal the diffusion of Si and O elements from the glass side into the stainless steel, along with traces of Fe and Mn elements penetrating into the glass side. These findings indicate the formation of a mixed molten pool during laser welding, which subsequently solidifies to form the mosaic structure. This intermixing is the fundamental reason for the successful bonding between the stainless steel and glass. Fig. 18(d) shows a higher-magnification (1000×) SEM image of the weld seam. The corresponding EDS mappings in Fig. 18(e) and (f) further clarify the elemental distributions. In Fig. 18(d), the mosaic structure becomes more distinct, and numerous white circular features can be observed on the glass side. Based on the elemental analysis in Fig. 18(e) and (f), these white features are identified as glass-metal spatters. These spatters are generated due to recoil pressure during welding, causing molten material to be ejected. Upon cooling, they remain embedded in the interface, and during the shear test, fractures occurring around these spatters result in the formation of spatter-induced pits. Additionally, irregularly shaped voids are also visible on the glass side, which likely result from thermal expansion mismatch between the two materials during re-solidification [19]. Fig. 18(h) displays the EDS line scan of the weld seam, with the specific scan position indicated by the yellow line in Fig. 18(g). The line scan profile in Fig. 18(h) shows a general trend in which Si and O contents decrease from the glass side to the stainless steel side, while Fe and Mn contents increase accordingly. Notably, sudden shifts in Si and Fe intensities, as highlighted by red boxes, indicate regions of material intermixing where the glass-metal molten pool was formed during welding. This elemental transition corresponds to the interfacial bonding layer. Finally, Fig. 18(i) presents an SEM image of a non-welded region, which was not exposed to the laser. The distinct three-layer structure comprising glass, mounting resin, and stainless steel can be clearly seen, confirming the absence of fusion in this region.



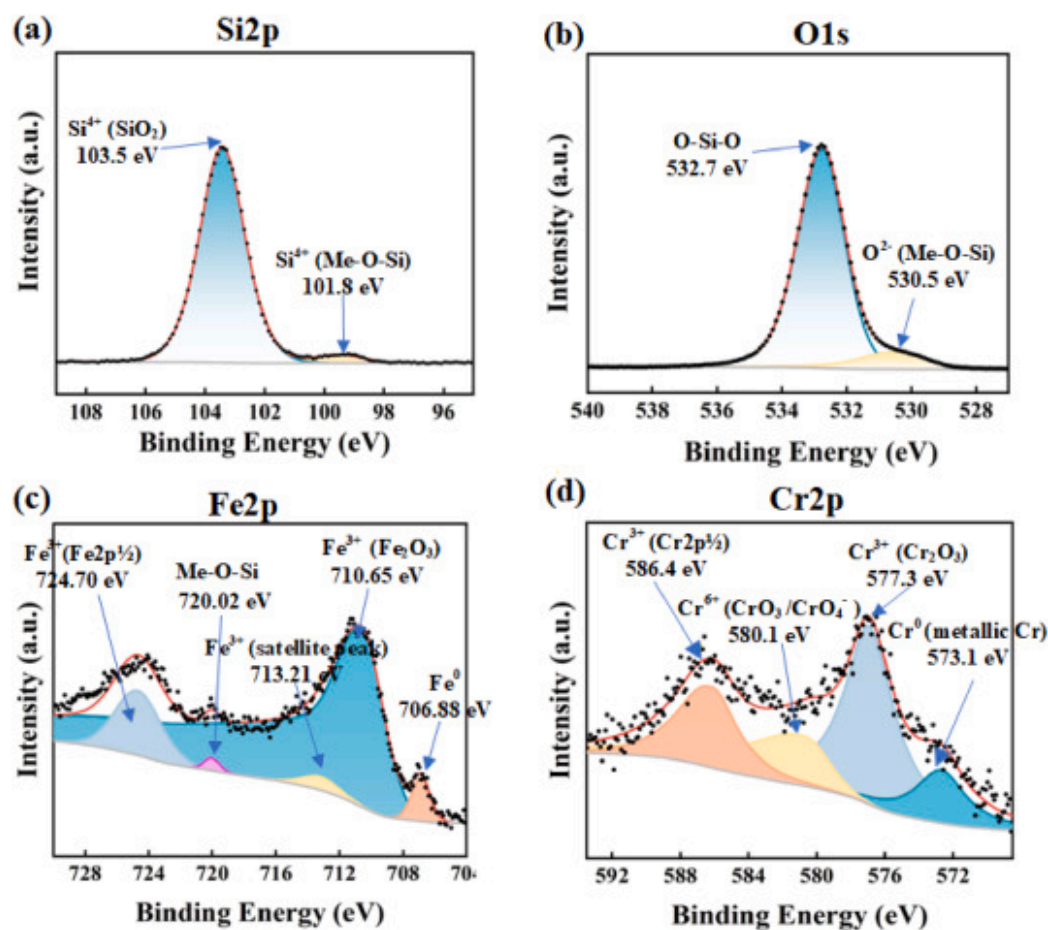
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Fig. 18. SEM and EDS images of the joint cross-section: (a) SEM image of the weld seam at 500×magnification; (b, c) EDS elemental maps corresponding to the area shown in (a); (d) SEM image of the weld seam at 1000×magnification; (e, f) EDS elemental maps corresponding to the area shown in (c); (g) Schematic indicating the EDS line scan region across the weld seam; (h) EDS line scan result corresponding to the line shown in (g); (i) SEM image of the non-welded region of the joint.

3.3. Chemical bonding mechanisms via XPS analysis

To further elucidate the chemical bonding mechanisms at the welded interface, high-resolution XPS analysis was performed on the fracture surface of the stainless steel side, as shown in Fig. 19(a)–(d).



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Fig. 19. High-resolution XPS spectra obtained from the stainless steel side of the fracture interface: (a) Si 2p, (b) O 1s, (c) Fe 2p, and (d) Cr 2p.

The Si 2p spectrum (Fig. 19(a)) exhibits a dominant peak at 103.5 eV corresponding to Si^{4+} in SiO_2 , and a secondary peak at 101.8 eV, which is indicative of silicon atoms involved in metal-oxygen-silicon (Me-O-Si) environments. The O 1s spectrum (Fig.

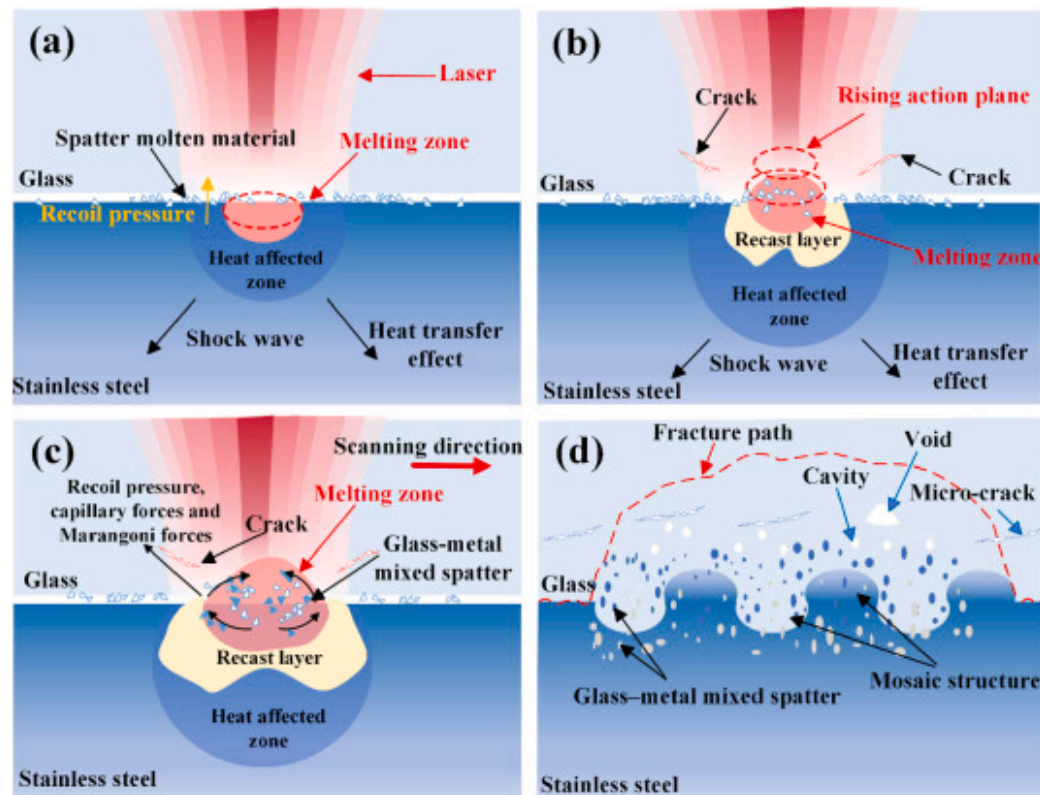
19(b)) contains two components: a peak at 532.7 eV associated with Si-O bonding, and a lower-energy peak at 530.5 eV attributed to O^{2-} in metal oxides or Me-O-Si structures. The Fe 2p spectrum (Fig. 19(c)) shows several features: the main $Fe^{3+} 2p_{3/2}$ peak at 710.65 eV (Fe_2O_3), a metallic Fe^0 peak at 706.88 eV, a satellite peak at 713.21 eV characteristic of Fe^{3+} , and notably, a shoulder at 720.02 eV. This latter peak does not correspond to conventional Fe species and is likely the result of a chemical shift due to Fe-O-Si bond formation at the interface. In the Cr 2p spectrum (Fig. 19(d)), peaks are observed at 577.3 eV and 586.4 eV, corresponding to Cr^{3+} (Cr_2O_3), and at 573.1 eV for metallic Cr^0 . A peak at 580.1 eV is consistent with Cr^{6+} species, such as CrO_3 or CrO_4^{2-} , likely formed due to localized high-temperature oxidation during laser welding. Although Cr^{6+} may participate in interface reactions, there is limited evidence to support the formation of Cr-O-Si bonds; Fe is more likely the dominant contributor in forming Me-O-Si linkages.

These XPS results provide compelling evidence for the formation of interfacial Me-O-Si bonds, especially Fe-O-Si, which act as atomic bridges between the stainless steel and quartz glass. The presence of such covalent bonding networks is well known to enhance interfacial strength in dissimilar laser-welded joints [43].

3.4. Proposed interfacial bonding mechanism and process model

Based on the fracture and cross-section morphology (Sections 3.1–3.2) as well as elemental analysis (Section 3.3), we now construct a comprehensive model to describe the physical and chemical processes during nanosecond LTW. Fig. 20 illustrates the schematic of the welding process between stainless steel and glass using a nanosecond pulsed laser. As shown in Fig. 20(a), during the initial stage, the laser emitted from the nanosecond pulsed laser system irradiates the stainless steel-glass interface. The laser energy rapidly heats the surface of the stainless steel, causing localized melting and the formation of a melting zone. At the same time, high temperatures lead to the generation of metal vapor, which produces a recoil pressure that drives molten stainless steel to splash [44]. Heat conduction transfers energy into the bulk of the material, forming a HAZ, while shock waves are also generated and propagate inward. In Fig. 20(b), as laser irradiation continues, the optical transparency of the glass diminishes and it begins to absorb laser energy. The melting zone expands upward, resulting in the further melting of glass, while the previously melting zone starts to solidify, forming a recast layer. Concurrently, due to localized thermal stress, cracks may develop in the glass near the interface. As shown in Fig. 20(c), thermal gradients at the weld interface may induce surface tension-driven convection, often referred to as the Marangoni effect [45]. While Marangoni flow is typically more pronounced in continuous wave or millisecond laser welding due to sustained thermal input, localized and transient thermal gradients during nanosecond pulsed laser irradiation may still contribute to micro-scale fluid motion within the molten pool, especially at

moderate repetition rates. Moreover, although the pulse duration (200 ns) and repetition frequency (up to 80kHz) limit the lifetime and scale of the molten pool, transient melt dynamics, driven by recoil pressure and localized thermal capillary forces, may synergistically influence the interfacial material distribution. This hypothesis aligns with observations in recent studies on pulsed laser welding of dissimilar materials [43], where even under sub-millisecond regimes, surface tension gradients contributed to melt shaping. Nevertheless, it is acknowledged that the Marangoni effect is not the sole or dominant driving mechanism under nanosecond laser regimes, and its contribution likely acts in concert with recoil pressure, capillary forces and transient vapor dynamics to affect molten pool behavior. During the final stage, as depicted in Fig. 20(d), the laser is turned off and the molten pool cools and solidifies. A mosaic structure forms at the stainless steel-glass interface, completing the welding process. Moreover, various defects such as glass-metal composite spatters, internal voids, cavities, and micro-cracks are observed within the weld seam. Throughout this process, the successful joining of glass and stainless steel results from the combined effects of laser energy input, thermal conduction, vapor-induced recoil pressure, capillary forces and Marangoni convection.



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Fig. 20. Schematic illustration of the welding process between stainless steel and glass using a nanosecond pulsed laser.

The observed interfacial bonding mechanisms across the weld seam suggest a sequential relationship between localized melting and mechanical interlocking, rather than a parallel or competing contribution. Localized melting is a prerequisite for the formation of mechanical interlocking structures. When the laser is focused near the glass-metal interface, sufficient thermal input leads to partial melting of both materials, followed by rapid solidification and interfacial mixing. This process generates various interlocking features, such as embedded spatters and rough interfacial geometries.

The smooth zone is typically formed in regions far from the laser focal point, where melting is most pronounced and the re-solidified structure is compact and continuous. In this region, interfacial bonding is dominated by cohesive fracture within the glass bulk, as evidenced by deep pits and uniform elemental mixing, leading to higher shear strength. In contrast, the fragment region is associated with regions receiving suboptimal thermal input, resulting in limited or incomplete melting. Here, only weak or sparse interlocking features are observed, and the fracture tends to propagate along the interface due to insufficient joint integrity. The fracture surfaces in this zone show flatter profiles with scattered micro-fragments, suggesting an interfacial adhesion failure mode [46,47].

In the present nanosecond laser welding experiments, the relatively large error bars in the measured shear strength (as shown in Fig. 4, Fig. 9, Fig. 12, Fig. 15) reflect the inherent variability of the process. Several factors contribute to this data dispersion. First, the interaction between the pulsed laser and the glass-metal interface generates plasma, which exhibits highly nonlinear feedback behavior. Small fluctuations in experimental conditions or sample properties can lead to significant variations in plasma density, distribution, or duration, resulting in inconsistent energy deposition due to plasma shielding [48]. Second, the considerable mismatch in thermal expansion coefficients between quartz glass and stainless steel promotes the formation of transverse microcracks and thermal stress accumulation near the weld, which may vary from sample to sample and affect mechanical performance. Third, slight inconsistencies in the surface roughness and optical contact of the bonded samples can cause fluctuations in laser coupling efficiency, further affecting weld uniformity [49]. These combined physical effects and experimental factors contribute to the observed scatter in shear strength results.

In addition to the physical bonding and elemental interdiffusion at the interface, chemical bonding also plays a critical role in strengthening the welded joint. As revealed by XPS analysis, the formation of Me-O-Si (primarily Fe-O-Si) bonds at the stainless

steel/quartz glass interface creates strong covalent bridges that enhance the mechanical interlocking at the atomic level. These chemical linkages contribute significantly to the improved shear strength observed in the welded joints, supplementing the physical fusion and metallurgical bonding mechanisms.

4. Conclusions

In this study, a nanosecond LTW approach was employed to join 304 stainless steel and quartz glass under optical contact conditions maintained by mechanical clamping. The effects of key laser parameters, including power, pulse frequency, scanning speed, and scanning times, were systematically investigated through single-factor experiments to understand their influence on joint morphology and bonding characteristics. The bonding mechanisms were further explored via mechanical testing, fracture surface analysis, and cross-sectional characterization using SEM and EDS. The main conclusions are summarized as follows.

1. A maximum shear strength of 21.39MPa was achieved at a laser power of 11.4W, frequency of 40kHz, scanning speed of 100mm/s, and 30 scanning times. The experimental results demonstrated that each parameter exhibited a process window within which favorable joint formation occurred. Excessive heat input led to thermal stress accumulation, over-ablation, and microcrack formation, whereas insufficient input caused poor molten pool formation and weak bonding.
2. SEM and EDS analyses of the fracture surfaces identified three distinct morphological regions in the weld area: smooth, fragmented, and ablated. The smooth region, which expanded under optimized conditions, was found to contribute most significantly to joint integrity. Fracture analysis showed that failure predominantly initiated within the glass, indicating cohesive failure and strong bonding at the interface.
3. Cross-sectional imaging and elemental mapping suggested that interfacial bonding was governed by a synergistic mechanism involving localized melting and subsequent mechanical interlocking. Localized melting at the glass-metal interface enabled elemental interdiffusion and the formation of a mixed molten pool, which upon rapid solidification generated interlocking microstructures. The extent and compactness of these features were directly associated with the degree of localized melting, particularly evident in the smooth zone where bonding strength was highest.

4. XPS analysis revealed the presence of interfacial chemical bonding in the form of Me-O-Si structures, particularly Fe-O-Si. These covalent bonds serve as atomic bridges between the stainless steel and quartz glass, enhancing the bonding strength beyond physical interlocking. The coexistence of chemical bonding and mechanical interlocking was identified as the dominant mechanism contributing to the high-strength interface.

The optimized parameter combination (11.4W, 40kHz, 100mm/s, 30 scanning times) shows potential for practical applications in dissimilar material joining, especially in optoelectronic or biomedical packaging where transparent-to-metal bonding with high strength is required.

CRediT authorship contribution statement

Runhan Zhao: Writing – original draft, Validation, Formal analysis, Conceptualization. **Yide Li:** Validation, Formal analysis, Data curation. **Xiangfeng Liu:** Validation, Formal analysis, Data curation. **Yongzhe Li:** Resources, Investigation, Funding acquisition. **Jiajun Fu:** Validation, Formal analysis, Data curation. **Houfu Dai:** Resources, Investigation. **Huixin Wang:** Resources, Investigation. **Tairui Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Qinghua Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is the Supplementary data to this article:

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Multimedia component 1.

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